

green

```
# ==== Manual configuration
setwd("/Users/luccahenrique/LSD/Work/green_made/green_model_v3") # set working directory

library(ggplot2)
library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr    1.5.1
## v lubridate  1.9.4      v tibble     3.2.1
## v purrr      1.0.4      v tidyr      1.3.1
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

library(mFilter)
library(ggcorrplot)
library(writexl)

library(knitr)
library(tidyr)
library(dplyr)
library(purrr)

load("baseline_aug25.RData")

load("scenario2_jul25.RData")

load("scenario3_jul25.RData")

load("scenario4_jul25.RData")

load("scenario5_jul25.RData")

load("scenario6_jul25_capitalgoods.RData")

load("scenario7_jul25.RData")

load("scenario8_jul25.RData")

load("scenario9_jul25.RData")

load("scenario10_jul25.RData")

load("scenario11_jul25_capitalgoods.RData")
```

```

df_baseline$periods <- 1:nrow(df_baseline)

df_scenario2$periods <- 1:nrow(df_scenario2)

df_scenario3$periods <- 1:nrow(df_scenario3)

df_scenario4$periods <- 1:nrow(df_scenario4)

df_scenario5$periods <- 1:nrow(df_scenario5)

df_scenario6$periods <- 1:nrow(df_scenario6)

df_scenario7$periods <- 1:nrow(df_scenario7)

df_scenario8$periods <- 1:nrow(df_scenario8)

df_scenario9$periods <- 1:nrow(df_scenario9)

df_scenario10$periods <- 1:nrow(df_scenario10)

df_scenario11$periods <- 1:nrow(df_scenario11)

# List of scenario data frames
scenarios_carbontax <- list(df_baseline, df_scenario2, df_scenario3, df_scenario4, df_scenario5, df_scenario6, df_scenario7, df_scenario8, df_scenario9, df_scenario10, df_scenario11)

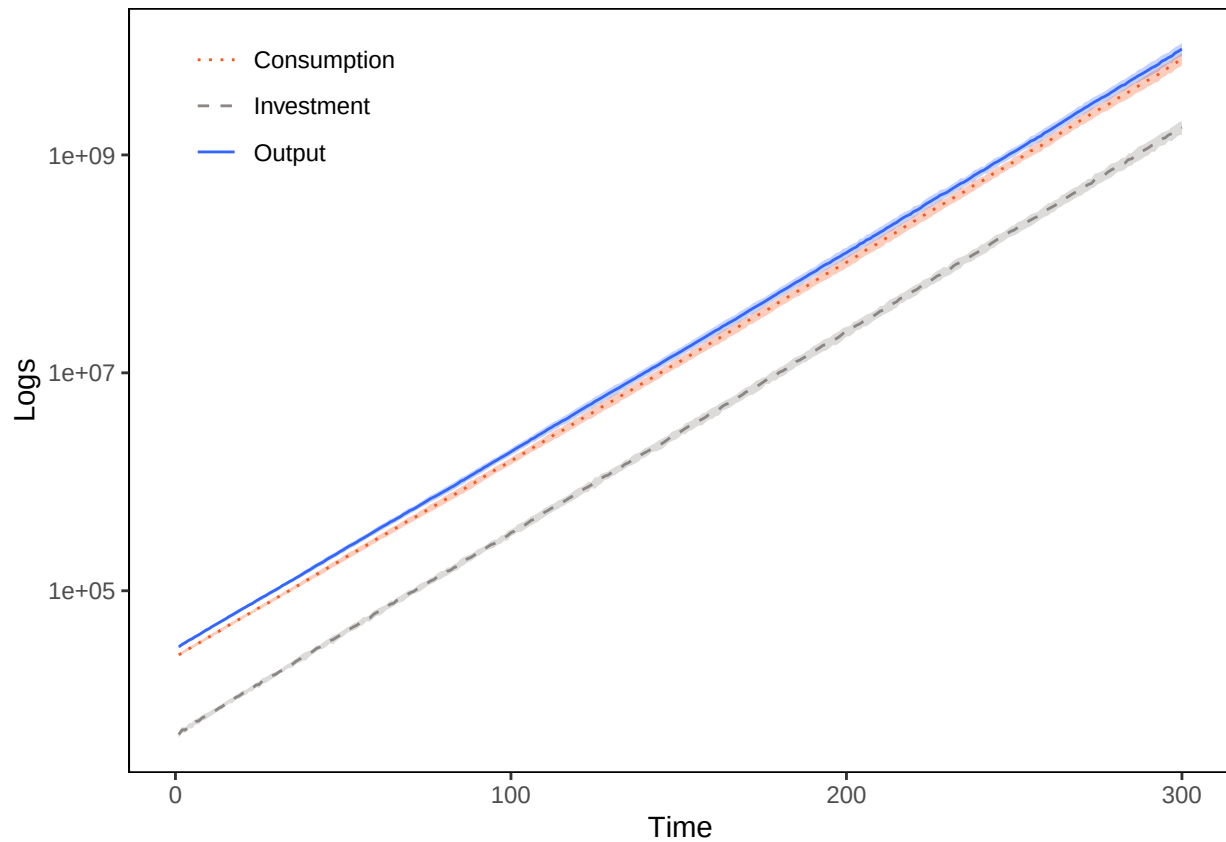
scenarios_wealthtax <- list(df_baseline, df_scenario7, df_scenario8, df_scenario9, df_scenario10, df_scenario11)

ggplot(df_baseline) +
  geom_line(aes(x = periods, y = gdp_real_Med, color = "Output")) +
  geom_ribbon(aes(x = periods, ymin = gdp_real_Med - gdp_real_Mad, ymax = gdp_real_Med + gdp_real_Mad, color = "Output")) +
  geom_line(aes(x = periods, y = investment_realized_c_Med, color = "Investment"), linetype = "dashed") +
  geom_ribbon(aes(x = periods, ymin = investment_realized_c_Med - investment_realized_c_Mad, ymax = investment_realized_c_Med + investment_realized_c_Mad, color = "Investment"), linetype = "dashed") +
  geom_line(aes(x = periods, y = production_c_real_Med, color = "Consumption"), alpha = 1.1, linetype = "solid") +
  geom_ribbon(aes(x = periods, ymin = production_c_real_Med - production_c_real_Mad, ymax = production_c_real_Med + production_c_real_Mad, color = "Consumption"), alpha = 1.1, linetype = "solid") +
  scale_y_continuous(trans='log10') +
  theme(legend.position = "bottom") +
  labs(x = "Time", y = "Logs", color = "") +
  theme(panel.border = element_rect(color = "black", linewidth = 0.5, fill = NA),
        panel.background = element_rect(fill = "white"),
        legend.key = element_blank(),
        legend.position = c(0.15, 0.9),
        legend.background = element_blank()) +
  scale_color_manual(values = c( "#f7581d", "#8b8683", "#3366ff" ),
                     guide = guide_legend(override.aes = list(linetype = c("dotted", "dashed", "solid")))) +
  scale_fill_manual(values = c( "#f7581d", "#8b8683", "#3366ff" ))

## Warning: A numeric `legend.position` argument in `theme()` was deprecated in ggplot2
## 3.5.0.
## i Please use the `legend.position.inside` argument of `theme()` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was

```

generated.



```
lowP      <- 6      # bandpass filter minimum period
highP     <- 32     # bandpass filter maximum period
bpfK      <- 12     # bandpass filter order

series_list <- list(
  gdp_real_Med = "GDP",
  production_c_real_Med = "Consumption",
  investment_realized_c_Med = "Investment",
  unemployment_rate_Med = "Unemployment",
  cpi_Med = "CPI",
  gini_hh_Med = "Gini",
  wage_share_Med = "Wage share",
  emissions_Med = "Emissions"
)

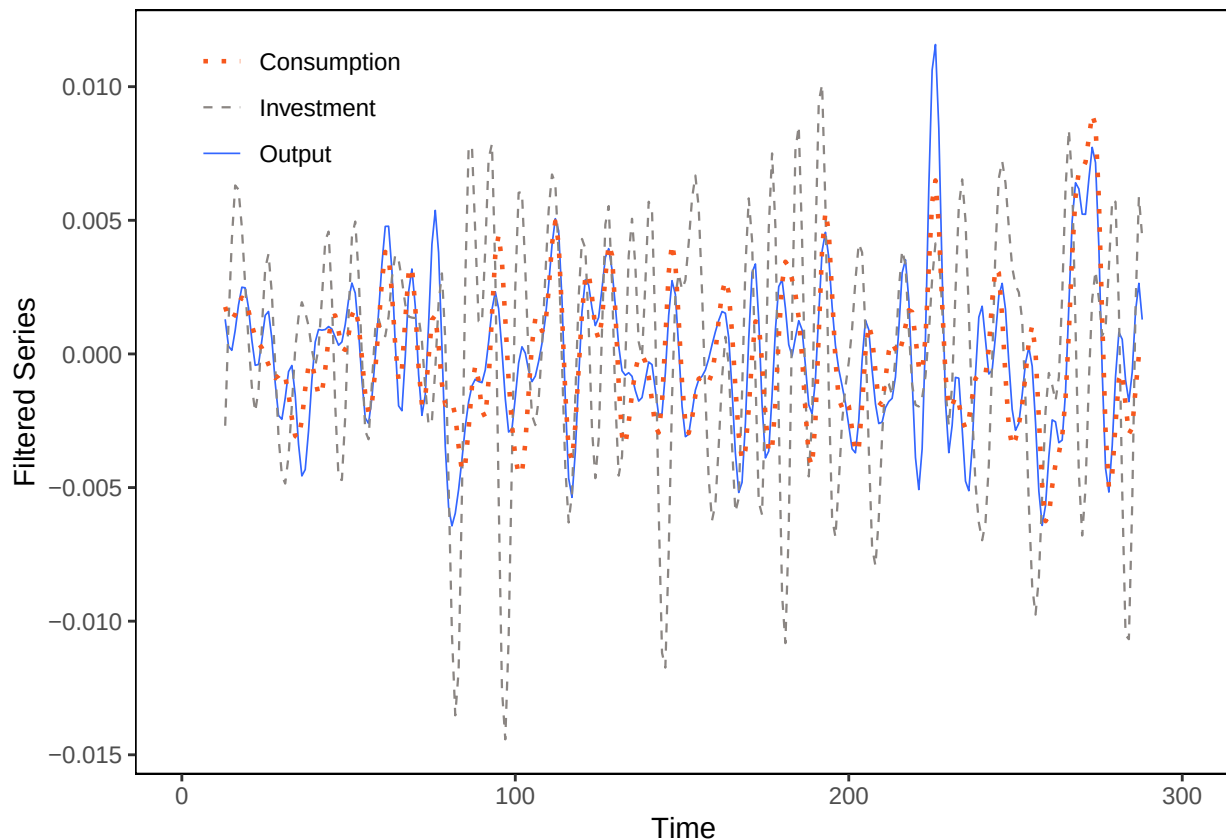
# Function to apply bkfilter and extract the cycle
extract_cycle <- function(series_name, new_name) {
  cycle <- bkfilter(log(df_baseline[[series_name]]), pl = lowP, pu = highP, nfix = bpfK)$cycle
  setNames(data.frame(cycle), new_name)
}

# Apply to all series and combine columns
cycles_combined <- map2_dfc(names(series_list), series_list, extract_cycle)
cycles_combined$periods <- 1:nrow(df_scenario11)
```

```

ggplot(data = cycles_combined) +
  geom_line(aes(x = periods, y = GDP, color = "Output"),linewidth = 0.3, na.rm = TRUE) +
  geom_line(aes(x = periods, y = Investment, color= "Investment"),linetype = "dashed", linewidth = 0.4,
  geom_line(aes(x = periods, y = Consumption, color = "Consumption"), linewidth = 0.8, alpha =1.2,linetype = "dotted") +
  theme(legend.position = "bottom") +
  labs(x = "Time", y= "Filtered Series", color = "") +
  theme(panel.border = element_rect(color = "black", linewidth = 0.5, fill = NA),
        panel.background = element_rect(fill = "white"),
        legend.key = element_blank(),
        legend.position = c(0.15,0.9),
        legend.background = element_blank()) +
  scale_color_manual(values = c( "#f7581d", "#8b8683" ,"#3366ff"),
                     guide = guide_legend(override.aes = list(linetype = c("dotted", "dashed", "solid"))))

```



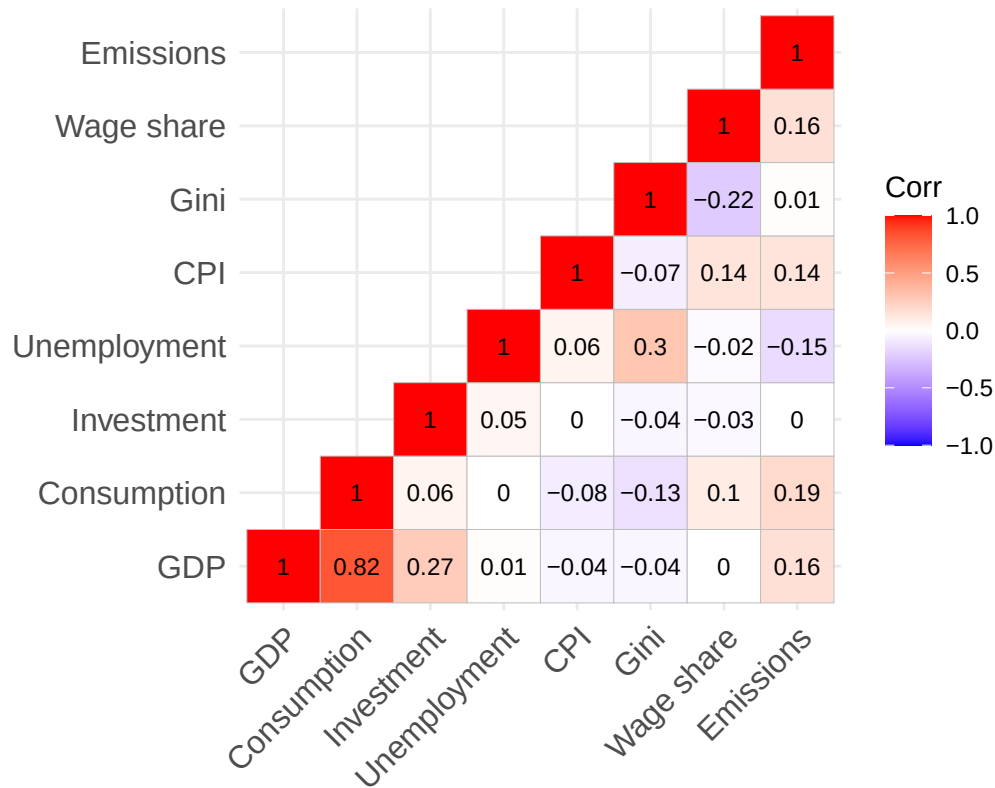
```

cor_matrix <- cor(cycles_combined[ , !names(cycles_combined) %in% "periods"],
                  use = "pairwise.complete.obs",
                  method = "pearson")

ggcorrplot(cor_matrix,
            type = "lower",           # or "full" to show both triangles
            lab = TRUE,              # show correlation values
            lab_size = 3,
            show.diag = TRUE,        # <- ensures diagonal is shown
            colors = c("blue", "white", "red"),
            title = "Pearson Correlation Matrix")

```

Pearson Correlation Matrix



```
# table_summary <- df_baseline %>%
# summarise(
#   gdp_mean = mean(gdp_growth_Med, na.rm = TRUE),
#   gdp_stdv = sd(gdp_growth_Med, na.rm = TRUE),
#   investment_mean = mean(investment_growth_Med, na.rm = TRUE),
#   investment_stdv = sd(investment_growth_Med, na.rm = TRUE),
#   consumption_mean = mean(consumption_growth_Med, na.rm = TRUE),
#   consumption_stdv = sd(consumption_growth_Med, na.rm = TRUE),
#   unemployment_mean = mean(unemployment_rate_Med, na.rm = TRUE),
#   unemployment_stdv = sd(unemployment_rate_Med, na.rm = TRUE),
# )

# List of variables to compute CCF with GDP
var_gdp_ccf <- c("Consumption", "Investment", "Unemployment", "Emissions")

# Function to compute CCF with GDP
get_ccf <- function(var_name) {
  ccf_res <- ccf(cycles_combined$GDP, cycles_combined[[var_name]], lag.max = 3, na.action = na.pass, plot = FALSE)
  data.frame(
    Variable = var_name,
    Lag = ccf_res$lag,
    CCF = ccf_res$acf
  )
}

# Combine results for all variables
```

```
ccf_table <- map_df(var_gdp_ccf, get_ccf)

# Pivot to wide format for LaTeX table
ccf_latex_table <- ccf_table %>%
  pivot_wider(names_from = Lag, values_from = CCF, names_prefix = "t")

# Optional: Round for presentation
ccf_latex_table <- ccf_latex_table %>%
  mutate(across(starts_with("t"), ~round(., 3)))

ccf_latex_table
```

A tibble: 4 x 8

Variable t-3 t-2 t-1 t0 t1 t2 t3 1 Consumption 0.274 0.557 0.777 0.822 0.661 0.375 0.096 2 Investment -0.135 -0.053 0.115 0.266 0.305 0.211 0.054 3 Unemployment -0.061 -0.093 -0.059 0.01 0.066 0.071 0.033 4 Emissions 0.099 0.169 0.189 0.161 0.102 0.046 0.017

```
# List of variables to compute CCF with GDP
var_gini_ccf <- c("GDP", "Investment", "Unemployment", "Wage share")

# Function to compute CCF with GDP
get_ccf <- function(var_name) {
  ccf_res <- ccf(cycles_combined$Gini, cycles_combined[[var_name]], lag.max = 3, na.action = na.pass, p
  data.frame(
    Variable = var_name,
    Lag = ccf_res$lag,
    CCF = ccf_res$acf
  )
}

# Combine results for all variables
ccf_table <- map_df(var_gini_ccf, get_ccf)

# Pivot to wide format for LaTeX table
ccf_latex_table <- ccf_table %>%
  pivot_wider(names_from = Lag, values_from = CCF, names_prefix = "t")

# Optional: Round for presentation
ccf_latex_table <- ccf_latex_table %>%
  mutate(across(starts_with("t"), ~round(., 3)))

ccf_latex_table
```

```
## # A tibble: 4 x 8
##   Variable   `t-3` `t-2` `t-1`   t0    t1    t2    t3
##   <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 GDP        0.074 0.067 0.027 -0.044 -0.127 -0.185 -0.18
## 2 Investment 0.001 0.043 0.013 -0.04  -0.058 -0.011 0.079
## 3 Unemployment 0.077 0.241 0.336 0.296 0.123 -0.087 -0.218
## 4 Wage share 0.041 0.003 -0.105 -0.221 -0.257 -0.181 -0.028
```

```
ggplot() +
  geom_line(data = df_baseline, aes(x = periods, y = gini_hh_Med, color = "Baseline"), alpha = 1) +
```

```

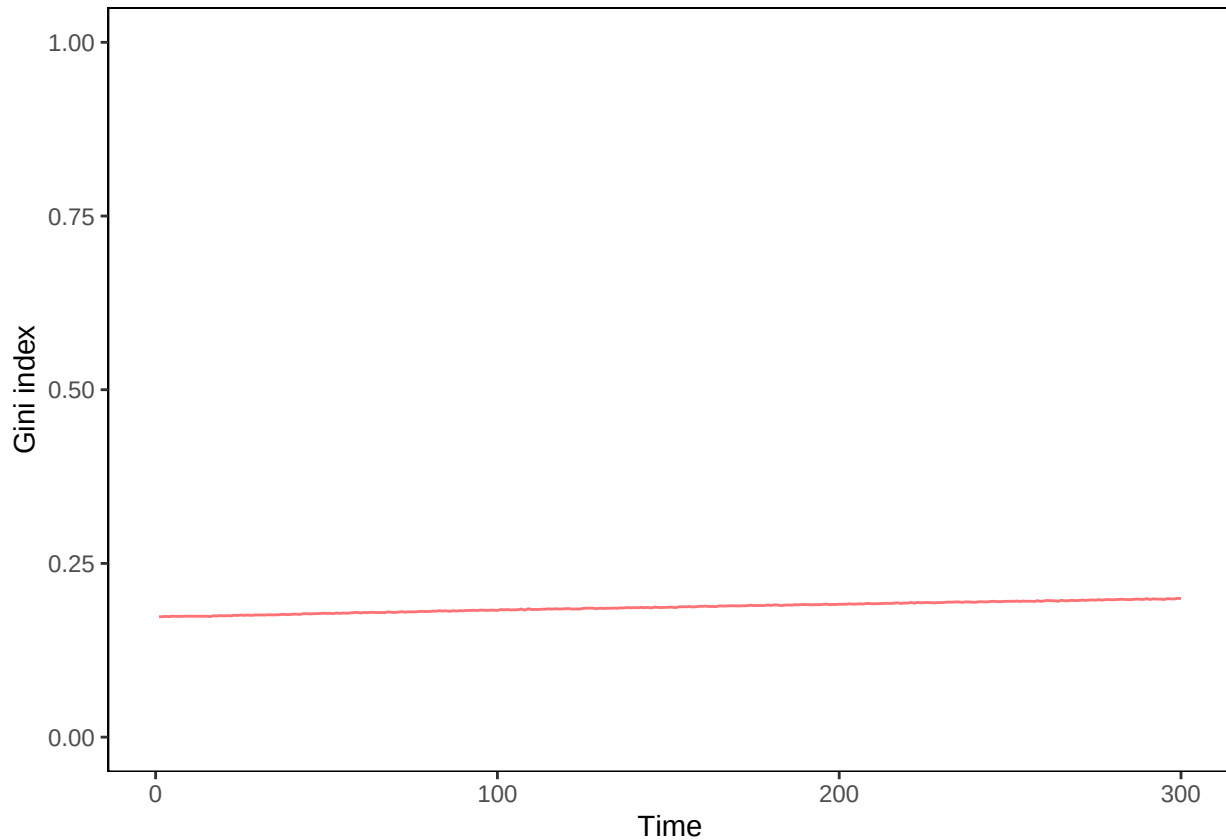
geom_ribbon(data = df_baseline, aes(x = periods, ymin = gini_hh_Med - 1.96 * gini_hh_Mad / sqrt(50),
                                   ymax = gini_hh_Med + 1.96 * gini_hh_Mad / sqrt(50), fill = "Baseline",
                                   alpha = 0.3, show.legend = FALSE) +
scale_color_manual(values = c("Baseline" = "#FF7276")) +
scale_fill_manual(values = c("Baseline" = "#FF7276")) +
labs(x = "Time", y = "Gini index", color = "", fill = "") +
theme(
  legend.position = "none",
  panel.border = element_rect(color = "black", size = 0.5, fill = NA),
  panel.background = element_rect(fill = "white")
)+
scale_y_continuous(limits = c(0, 1))

```

```

## Warning: The `size` argument of `element_rect()` is deprecated as of ggplot2 3.4.0.
## i Please use the `linewidth` argument instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

```



```

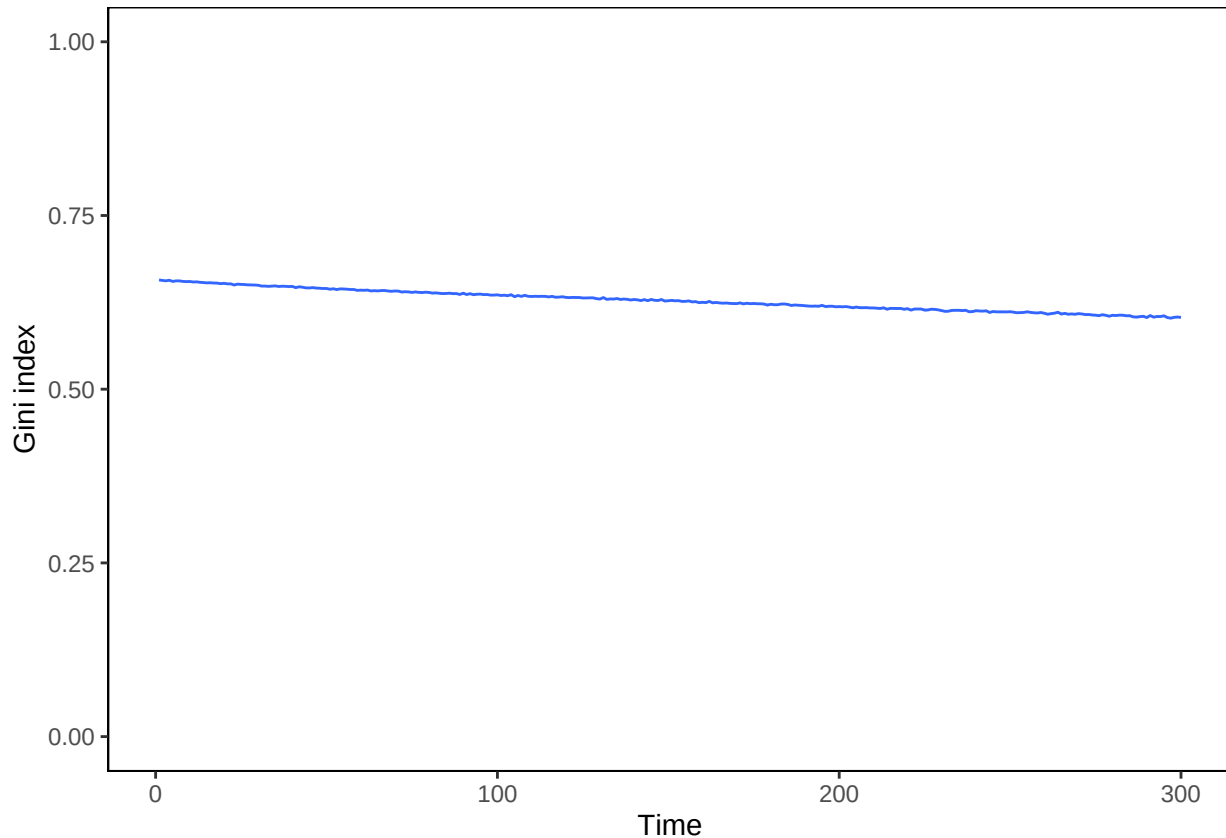
ggplot() +
geom_line(data = df_baseline, aes(x = periods, y = wage_share_Med, color = "Baseline"), alpha = 1) +
geom_ribbon(data = df_baseline, aes(x = periods, ymin = wage_share_Med - 1.96 * wage_share_Mad / sqrt(50),
                                   ymax = wage_share_Med + 1.96 * wage_share_Mad / sqrt(50), fill = "Baseline",
                                   alpha = 0.3, show.legend = FALSE) +
scale_color_manual(values = c("Baseline" = "#3366ff")) +
scale_fill_manual(values = c("Baseline" = "#3366ff")) +
labs(x = "Time", y = "Gini index", color = "", fill = "") +

```

```

theme(
  legend.position = "none",
  panel.border = element_rect(color = "black", size = 0.5, fill = NA),
  panel.background = element_rect(fill = "white")
) +
scale_y_continuous(limits = c(0, 1))

```



```

# Apply the calculations to each scenario
scenarios_carbontax <- lapply(scenarios_carbontax, function(df) {
  df$ratio_emissions_coefficient <- df$emissions_coefficient_Med / df$baseline$emissions_coefficient_Med

  df$corr_scenario <- cor(df$emissions_coefficient_Med, df$baseline$emissions_coefficient_Med)

  df$ratio_emissions_coefficient_sd <- abs(df$ratio_emissions_coefficient) * sqrt(
    (df$emissions_coefficient_Mad / df$emissions_coefficient_Med)^2 +
    (df$baseline$emissions_coefficient_Mad / df$baseline$emissions_coefficient_Med)^2 -
    2 * df$corr_scenario * (df$emissions_coefficient_Mad / df$emissions_coefficient_Med) * (df$baseline$emissions_coefficient_Mad / df$baseline$emissions_coefficient_Med)
  )

  df$ratio_emissions_coefficient_ci <- 1.96 * (df$ratio_emissions_coefficient_sd / sqrt(50))

  return(df)
})

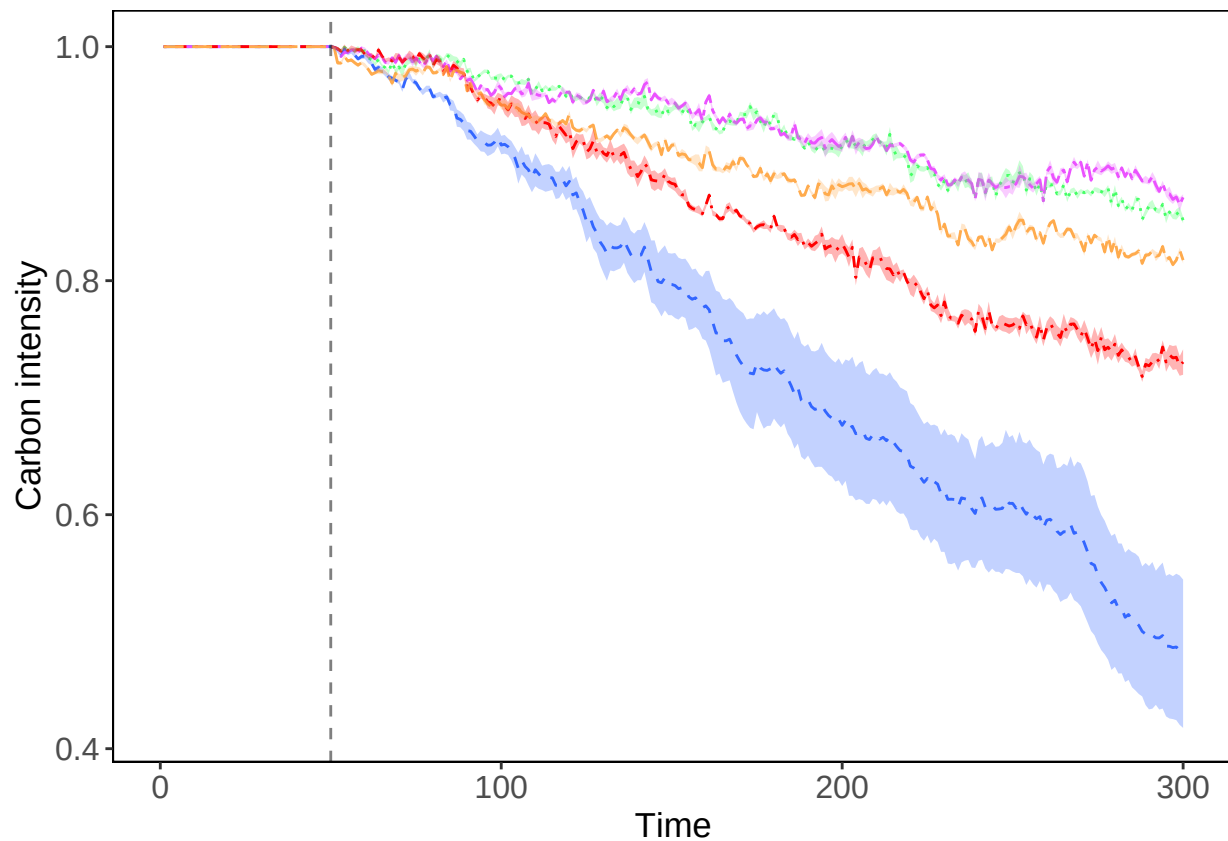
# Assign the updated data frames back to their respective variables
df_baseline <- scenarios_carbontax[[1]]
df_scenario2 <- scenarios_carbontax[[2]]

```



```
df_scenario3 <- scenarios_carbontax[[3]]
df_scenario4 <- scenarios_carbontax[[4]]
df_scenario5 <- scenarios_carbontax[[5]]
df_scenario6 <- scenarios_carbontax[[6]]

ggplot() +
  geom_line(data = df_scenario2, aes(x = periods, y = ratio_emissions_coefficient, color = "Carbon Tax"),
            data2 = df_scenario3, aes(x = periods, y = ratio_emissions_coefficient, color = "Carbon Tax"),
            data3 = df_scenario4, aes(x = periods, y = ratio_emissions_coefficient, color = "Carbon Tax"),
            data4 = df_scenario5, aes(x = periods, y = ratio_emissions_coefficient, color = "Carbon Tax"),
            data5 = df_scenario6, aes(x = periods, y = ratio_emissions_coefficient, color = "Carbon Tax")) +
  geom_ribbon(data = subset(df_scenario2, periods >= 50), aes(x = periods, ymin = ratio_emissions_coefficient, ymax = ratio_emissions_coefficient, color = "Carbon Tax"),
             data2 = subset(df_scenario3, periods >= 50), aes(x = periods, ymin = ratio_emissions_coefficient, ymax = ratio_emissions_coefficient, color = "Carbon Tax"),
             data3 = subset(df_scenario4, periods >= 50), aes(x = periods, ymin = ratio_emissions_coefficient, ymax = ratio_emissions_coefficient, color = "Carbon Tax"),
             data4 = subset(df_scenario5, periods >= 50), aes(x = periods, ymin = ratio_emissions_coefficient, ymax = ratio_emissions_coefficient, color = "Carbon Tax"),
             data5 = subset(df_scenario6, periods >= 50), aes(x = periods, ymin = ratio_emissions_coefficient, ymax = ratio_emissions_coefficient, color = "Carbon Tax")) +
  theme(legend.position = "bottom") +
  labs(x = "Time", y = "Carbon intensity", color = "", fill = "") +
  theme(panel.border = element_rect(color = "black", size = 0.5, fill = NA),
        axis.title.x = element_text(size = 13),
        axis.title.y = element_text(size = 13),
        axis.text.x = element_text(size = 12),
        axis.text.y = element_text(size = 12),
        panel.background = element_rect(fill = "white"),
        legend.key = element_blank(),
        legend.position = "none",
        legend.background = element_blank()) +
  scale_color_manual(values = c("#45ff66", "#EB52ff", "#3366ff", "#FFAB4D", "#ff0000"),
                     guide = guide_legend(override.aes = list(linetype = c("dotted", "dashed", "twodash")))) +
  scale_fill_manual(values = c("#45ff66", "#EB52ff", "#3366ff", "#FFAB4D", "#ff0000")) +
  geom_vline(xintercept = 50, linetype = "dashed", alpha = 0.5)
```



```
# Apply the calculations to each scenario
scenarios_carbontax <- lapply(scenarios_carbontax, function(df) {
  df$ratio_gini_hh <- df$gini_hh_Med / df_baseline$gini_hh_Med

  df$corr_scenario <- cor(df$gini_hh_Med, df_baseline$gini_hh_Med)

  df$ratio_gini_hh_sd <- abs(df$ratio_gini_hh) * sqrt(
    (df$gini_hh_Mad / df$gini_hh_Med)^2 +
    (df_baseline$gini_hh_Mad / df_baseline$gini_hh_Med)^2 -
    2 * df$corr_scenario * (df$gini_hh_Mad / df$gini_hh_Med) * (df_baseline$gini_hh_Mad / df_baseline$gini_hh_Med)
  )

  df$ratio_gini_hh_ci <- 1.96 * (df$ratio_gini_hh_sd / sqrt(50))

  return(df)
})

# Assign the updated data frames back to their respective variables
df_baseline <- scenarios_carbontax[[1]]
df_scenario2 <- scenarios_carbontax[[2]]
df_scenario3 <- scenarios_carbontax[[3]]
df_scenario4 <- scenarios_carbontax[[4]]
df_scenario5 <- scenarios_carbontax[[5]]
df_scenario6 <- scenarios_carbontax[[6]]

ggplot() +
```

```
geom_line(data = df_scenario2, aes(x = periods, y = ratio_gini_hh, color = "Carbon Tax"), linetype = "dotted"),
geom_line(data = df_scenario3, aes(x = periods, y = ratio_gini_hh, color = "Carbon Tax and green public goods"), linetype = "dotted"),
geom_line(data = df_scenario4, aes(x = periods, y = ratio_gini_hh, color = "Carbon Tax and green price incentives"), linetype = "dotted"),
geom_line(data = df_scenario5, aes(x = periods, y = ratio_gini_hh, color = "Carbon Tax and green research and development"), linetype = "dotted"),
geom_line(data = df_scenario6, aes(x = periods, y = ratio_gini_hh, color = "Carbon Tax and green public infrastructure"), linetype = "dotted"),
geom_ribbon(data = subset(df_scenario2, periods >= 50), aes(x = periods, ymin = ratio_gini_hh - ratio_gini_hh_min, ymax = ratio_gini_hh + ratio_gini_hh_max, color = "Carbon Tax"), linetype = "dotted"),
geom_ribbon(data = subset(df_scenario3, periods >= 50), aes(x = periods, ymin = ratio_gini_hh - ratio_gini_hh_min, ymax = ratio_gini_hh + ratio_gini_hh_max, color = "Carbon Tax and green public goods"), linetype = "dotted"),
geom_ribbon(data = subset(df_scenario4, periods >= 50), aes(x = periods, ymin = ratio_gini_hh - ratio_gini_hh_min, ymax = ratio_gini_hh + ratio_gini_hh_max, color = "Carbon Tax and green price incentives"), linetype = "dotted"),
geom_ribbon(data = subset(df_scenario5, periods >= 50), aes(x = periods, ymin = ratio_gini_hh - ratio_gini_hh_min, ymax = ratio_gini_hh + ratio_gini_hh_max, color = "Carbon Tax and green research and development"), linetype = "dotted"),
geom_ribbon(data = subset(df_scenario6, periods >= 50), aes(x = periods, ymin = ratio_gini_hh - ratio_gini_hh_min, ymax = ratio_gini_hh + ratio_gini_hh_max, color = "Carbon Tax and green public infrastructure"), linetype = "dotted"),
labs(x = "Time", y = "Gini index", color = "", fill = "") +
theme(panel.border = element_rect(color = "black", size = 0.5, fill = NA),
      axis.title.x = element_text(size = 13),
      axis.title.y = element_text(size = 13),
      axis.text.x = element_text(size = 12),
      axis.text.y = element_text(size = 12),
      panel.background = element_rect(fill = "white"),
      legend.key = element_blank(),
      legend.position = "none",
      legend.background = element_blank()) +
scale_color_manual(values = c("#45ff66", "#EB52ff", "#3366ff", "#FFAB4D", "#ff0000"),
                   guide = guide_legend(override.aes = list(linetype = c("dotted", "dashed", "twodash", "solid")))) +
scale_fill_manual(values = c("#45ff66", "#EB52ff", "#3366ff", "#FFAB4D", "#ff0000")) +
geom_vline(xintercept = 50, linetype = "dashed", alpha = 0.5)
```

```
# Apply the calculations to each scenario
scenarios_carbontax <- lapply(scenarios_carbontax, function(df) {
```

```

df$ratio_wage_share <- df$wage_share_Med / df_baseline$wage_share_Med

df$corr_scenario <- cor(df$wage_share_Med, df_baseline$wage_share_Med)

df$ratio_wage_share_sd <- abs(df$ratio_wage_share) * sqrt(
  (df$wage_share_Med / df$wage_share_Med)^2 +
  (df_baseline$wage_share_Med / df_baseline$wage_share_Med)^2 -
  2 * df$corr_scenario * (df$wage_share_Med / df$wage_share_Med) * (df_baseline$wage_share_Med / df_b
)

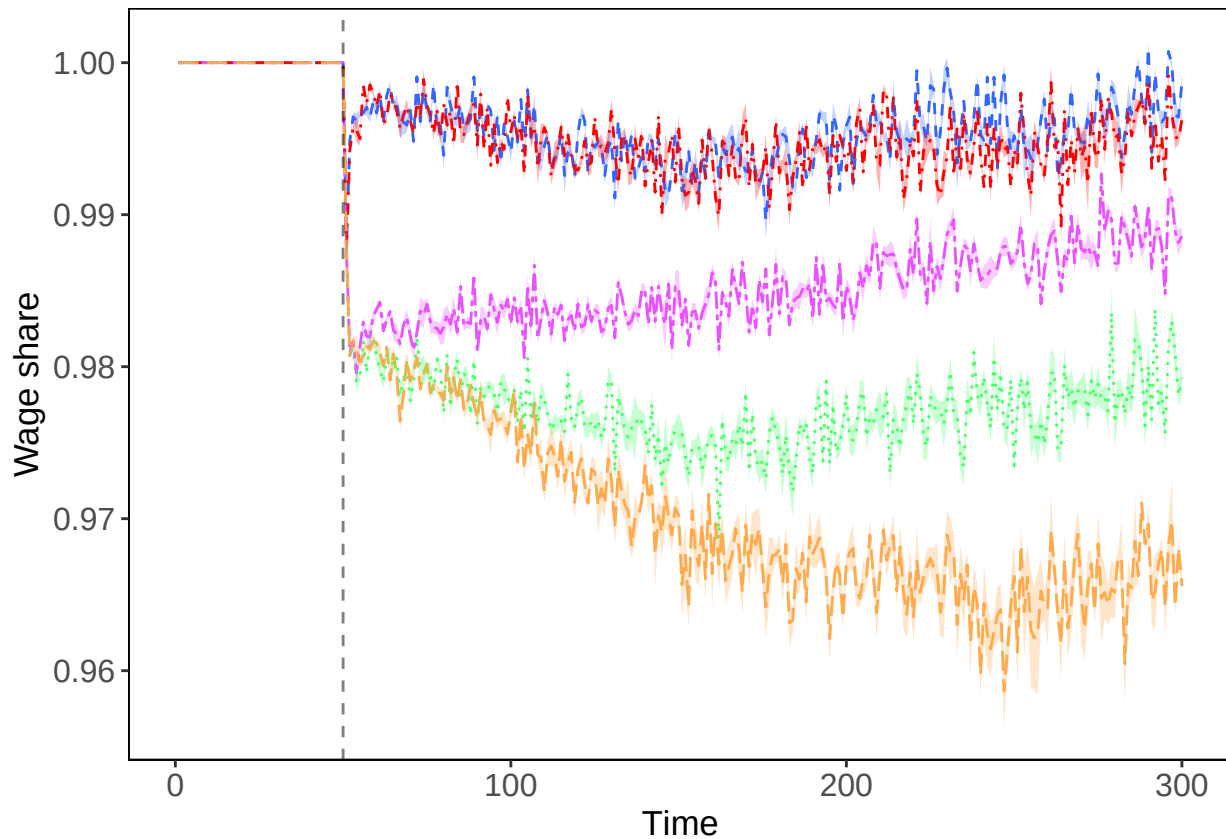
df$ratio_wage_share_ci <- 1.96 * (df$ratio_wage_share_sd / sqrt(50))

return(df)
})

# Assign the updated data frames back to their respective variables
df_baseline <- scenarios_carbontax[[1]]
df_scenario2 <- scenarios_carbontax[[2]]
df_scenario3 <- scenarios_carbontax[[3]]
df_scenario4 <- scenarios_carbontax[[4]]
df_scenario5 <- scenarios_carbontax[[5]]
df_scenario6 <- scenarios_carbontax[[6]]

ggplot() +
  geom_line(data = df_scenario2, aes(x = periods, y = ratio_wage_share, color = "Carbon Tax"), linetype
  geom_line(data = df_scenario3, aes(x = periods, y = ratio_wage_share, color = "Carbon Tax and green p
  geom_line(data = df_scenario4, aes(x = periods, y = ratio_wage_share, color = "Carbon Tax and green p
  geom_line(data = df_scenario5, aes(x = periods, y = ratio_wage_share, color = "Carbon Tax and green r
  geom_line(data = df_scenario6, aes(x = periods, y = ratio_wage_share, color = "Carbon Tax and green p
  geom_ribbon(data = subset(df_scenario2, periods >=50), aes(x = periods, ymin = ratio_wage_share - rat
  geom_ribbon(data = subset(df_scenario3, periods >=50), aes(x = periods, ymin = ratio_wage_share - rat
  geom_ribbon(data = subset(df_scenario4, periods >=50), aes(x = periods, ymin = ratio_wage_share - rat
  geom_ribbon(data = subset(df_scenario5, periods >=50), aes(x = periods, ymin = ratio_wage_share - rat
  geom_ribbon(data = subset(df_scenario6, periods >=50), aes(x = periods, ymin = ratio_wage_share - rat
  theme(legend.position = "bottom") +
  labs(x = "Time", y = "Wage share", color = "", fill = "") +
  theme(panel.border = element_rect(color = "black", size = 0.5, fill = NA),
        axis.title.x = element_text(size = 13),
        axis.title.y = element_text(size = 13),
        axis.text.x = element_text(size = 12),
        axis.text.y = element_text(size = 12),
        panel.background = element_rect(fill = "white"),
        legend.key = element_blank(),
        legend.position = "none",
        legend.background = element_blank()) +
  scale_color_manual(values = c("#45ff66", "#EB52ff", "#3366ff", "#FFAB4D", "#ff0000"),
                     guide = guide_legend(override.aes = list(linetype = c("dotted", "dashed", "twodash
  scale_fill_manual(values = c("#45ff66", "#EB52ff", "#3366ff", "#FFAB4D", "#ff0000")) +
  geom_vline(xintercept = 50, linetype = "dashed", alpha = 0.5)

```



```

# Apply the calculations to each scenario
scenarios_carbontax <- lapply(scenarios_carbontax, function(df) {
  df$ratio_gdp_real <- df$gdp_real_Med / df_baseline$gdp_real_Med

  df$corr_scenario <- cor(df$gdp_real_Med, df_baseline$gdp_real_Med)

  df$ratio_gdp_real_sd <- abs(df$ratio_gdp_real) * sqrt(
    (df$gdp_real_Med / df$gdp_real_Med)^2 +
    (df_baseline$gdp_real_Med / df_baseline$gdp_real_Med)^2 -
    2 * df$corr_scenario * (df$gdp_real_Med / df$gdp_real_Med) * (df_baseline$gdp_real_Med / df_baseline$gdp_real_Med)
  )

  df$ratio_gdp_real_ci <- 1.96 * (df$ratio_gdp_real_sd / sqrt(50))

  return(df)
})

# Assign the updated data frames back to their respective variables
df_baseline <- scenarios_carbontax[[1]]
df_scenario2 <- scenarios_carbontax[[2]]
df_scenario3 <- scenarios_carbontax[[3]]
df_scenario4 <- scenarios_carbontax[[4]]
df_scenario5 <- scenarios_carbontax[[5]]
df_scenario6 <- scenarios_carbontax[[6]]

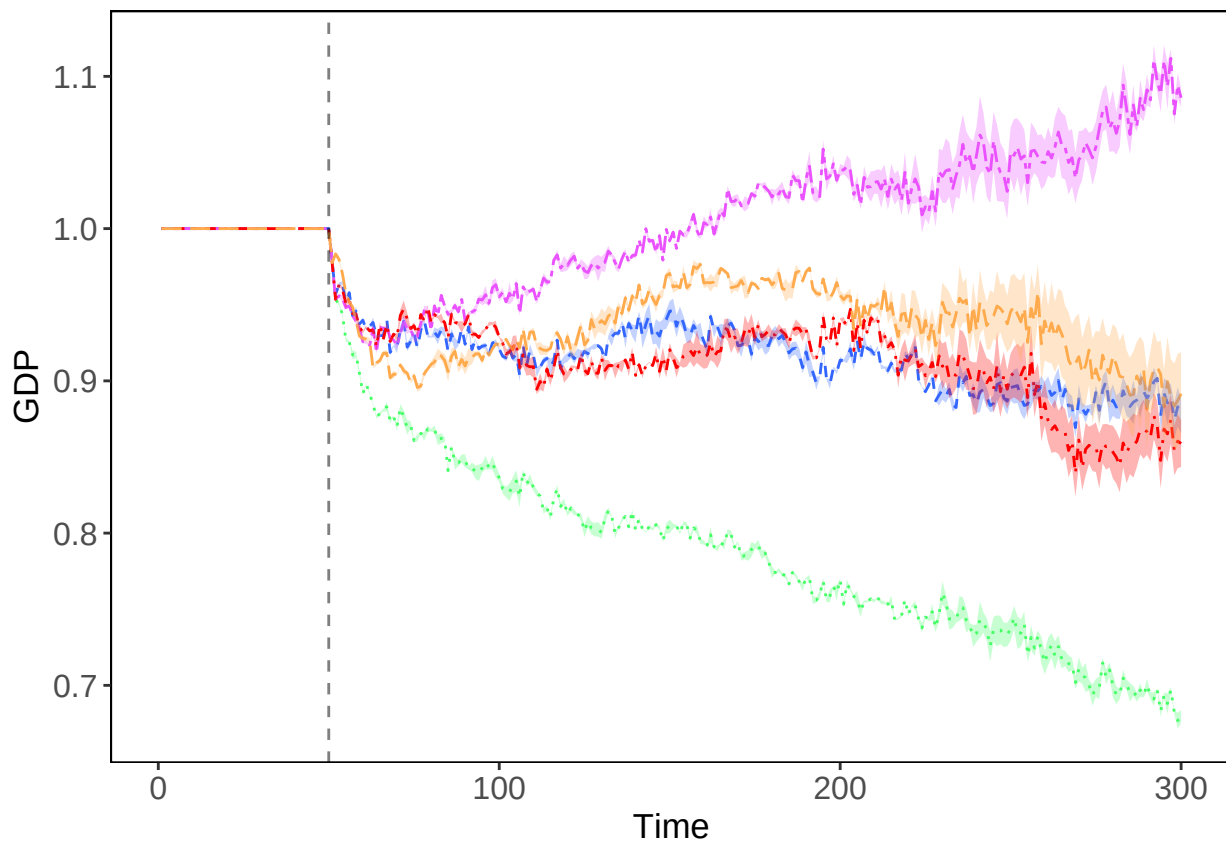
ggplot() +

```

```

geom_line(data = df_scenario2, aes(x = periods, y = ratio_gdp_real, color = "Carbon Tax"), linetype = 
geom_line(data = df_scenario3, aes(x = periods, y = ratio_gdp_real, color = "Carbon Tax and green pub
geom_line(data = df_scenario4, aes(x = periods, y = ratio_gdp_real, color = "Carbon Tax and green pri
geom_line(data = df_scenario5, aes(x = periods, y = ratio_gdp_real, color = "Carbon Tax and green res
geom_line(data = df_scenario6, aes(x = periods, y = ratio_gdp_real, color = "Carbon Tax and green pub
geom_ribbon(data = subset(df_scenario2, periods >= 50), aes(x = periods, ymin = ratio_gdp_real - ratio
geom_ribbon(data = subset(df_scenario3, periods >= 50), aes(x = periods, ymin = ratio_gdp_real - ratio
geom_ribbon(data = subset(df_scenario4, periods >= 50), aes(x = periods, ymin = ratio_gdp_real - ratio
geom_ribbon(data = subset(df_scenario5, periods >= 50), aes(x = periods, ymin = ratio_gdp_real - ratio
geom_ribbon(data = subset(df_scenario6, periods >= 50), aes(x = periods, ymin = ratio_gdp_real - ratio
theme(legend.position = "bottom") +
labs(x = "Time", y = "GDP", color = "", fill = "") +
theme(panel.border = element_rect(color = "black", size = 0.5, fill = NA),
      axis.title.x = element_text(size = 13),
      axis.title.y = element_text(size = 13),
      axis.text.x = element_text(size = 12),
      axis.text.y = element_text(size = 12),
      panel.background = element_rect(fill = "white"),
      legend.key = element_blank(),
      legend.position = "none",
      legend.background = element_blank()) +
scale_color_manual(values = c("#45ff66", "#EB52ff", "#3366ff", "#FFAB4D", "#ff0000"),
                   guide = guide_legend(override.aes = list(linetype = c("dotted", "dashed", "twodash
scale_fill_manual(values = c("#45ff66", "#EB52ff", "#3366ff", "#FFAB4D", "#ff0000")) +
geom_vline(xintercept = 50, linetype = "dashed", alpha = 0.5)

```



```

# Apply the calculations to each scenario
scenarios_carbontax <- lapply(scenarios_carbontax, function(df) {
  df$ratio_emissions <- df$emissions_Med / df_baseline$emissions_Med

  df$corr_scenario <- cor(df$emissions_Med, df_baseline$emissions_Med)

  df$ratio_emissions_sd <- abs(df$ratio_emissions) * sqrt(
    (df$emissions_Med / df$emissions_Med)^2 +
    (df_baseline$emissions_Med / df_baseline$emissions_Med)^2 -
    2 * df$corr_scenario * (df$emissions_Med / df$emissions_Med) * (df_baseline$emissions_Med / df_base
  )

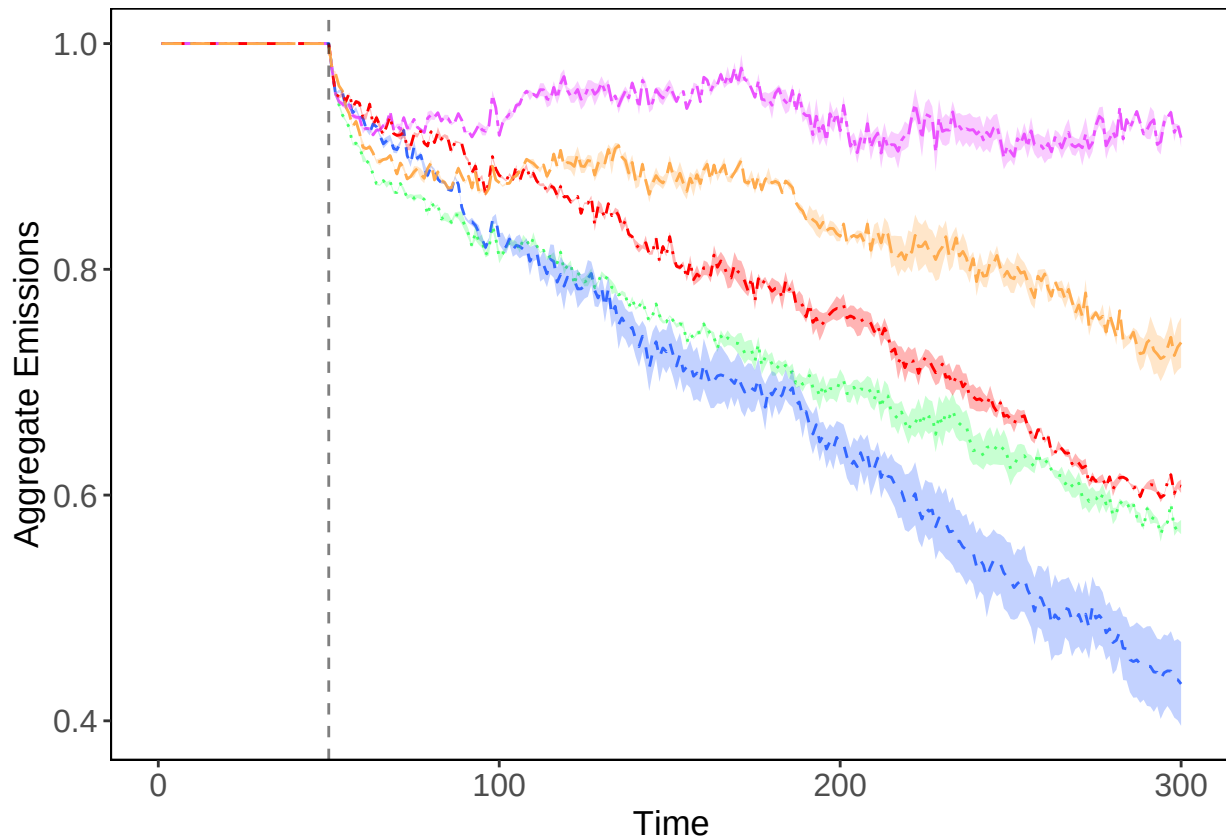
  df$ratio_emissions_ci <- 1.96 * (df$ratio_emissions_sd / sqrt(50))

  return(df)
})

# Assign the updated data frames back to their respective variables
df_baseline <- scenarios_carbontax[[1]]
df_scenario2 <- scenarios_carbontax[[2]]
df_scenario3 <- scenarios_carbontax[[3]]
df_scenario4 <- scenarios_carbontax[[4]]
df_scenario5 <- scenarios_carbontax[[5]]
df_scenario6 <- scenarios_carbontax[[6]]

ggplot() +
  geom_line(data = df_scenario2, aes(x = periods, y = ratio_emissions, color = "Carbon Tax"), linetype = "dotted") +
  geom_line(data = df_scenario3, aes(x = periods, y = ratio_emissions, color = "Carbon Tax and green pul"), linetype = "dashed") +
  geom_line(data = df_scenario4, aes(x = periods, y = ratio_emissions, color = "Carbon Tax and green pr"), linetype = "twodash") +
  geom_line(data = df_scenario5, aes(x = periods, y = ratio_emissions, color = "Carbon Tax and green re"), linetype = "solid") +
  geom_line(data = df_scenario6, aes(x = periods, y = ratio_emissions, color = "Carbon Tax and green pul"), linetype = "dotted") +
  geom_ribbon(data = subset(df_scenario2, periods >= 50), aes(x = periods, ymin = ratio_emissions - ratio_emissions_ci, ymax = ratio_emissions + ratio_emissions_ci, color = "Carbon Tax"), fill = "#45ff66", linetype = "dotted") +
  geom_ribbon(data = subset(df_scenario3, periods >= 50), aes(x = periods, ymin = ratio_emissions - ratio_emissions_ci, ymax = ratio_emissions + ratio_emissions_ci, color = "Carbon Tax and green pul"), fill = "#EB52ff", linetype = "dashed") +
  geom_ribbon(data = subset(df_scenario4, periods >= 50), aes(x = periods, ymin = ratio_emissions - ratio_emissions_ci, ymax = ratio_emissions + ratio_emissions_ci, color = "Carbon Tax and green pr"), fill = "#3366ff", linetype = "twodash") +
  geom_ribbon(data = subset(df_scenario5, periods >= 50), aes(x = periods, ymin = ratio_emissions - ratio_emissions_ci, ymax = ratio_emissions + ratio_emissions_ci, color = "Carbon Tax and green re"), fill = "#FFAB4D", linetype = "solid") +
  geom_ribbon(data = subset(df_scenario6, periods >= 50), aes(x = periods, ymin = ratio_emissions - ratio_emissions_ci, ymax = ratio_emissions + ratio_emissions_ci, color = "Carbon Tax and green pul"), fill = "#ff0000", linetype = "dotted") +
  theme(legend.position = "bottom") +
  labs(x = "Time", y = "Aggregate Emissions", color = "", fill = "") +
  theme(panel.border = element_rect(color = "black", size = 0.5, fill = NA),
        axis.title.x = element_text(size = 13),
        axis.title.y = element_text(size = 13),
        axis.text.x = element_text(size = 12),
        axis.text.y = element_text(size = 12),
        panel.background = element_rect(fill = "white"),
        legend.key = element_blank(),
        legend.position = "none",
        legend.background = element_blank()) +
  scale_color_manual(values = c("#45ff66", "#EB52ff", "#3366ff", "#FFAB4D", "#ff0000"),
                     guide = guide_legend(override.aes = list(linetype = c("dotted", "dashed", "twodash", "solid", "dotted")))) +
  scale_fill_manual(values = c("#45ff66", "#EB52ff", "#3366ff", "#FFAB4D", "#ff0000")) +
  geom_vline(xintercept = 50, linetype = "dashed", alpha = 0.5)

```

```
# Apply the calculations to each scenario
scenarios_carbontax <- lapply(scenarios_carbontax, function(df) {
  df$ratio_unemployment_rate <- df$unemployment_rate_Med / df_baseline$unemployment_rate_Med

  df$corr_scenario <- cor(df$unemployment_rate_Med, df_baseline$unemployment_rate_Med)

  df$ratio_unemployment_rate_sd <- abs(df$ratio_unemployment_rate) * sqrt(
    (df$unemployment_rate_Med / df$unemployment_rate_Med)^2 +
    (df_baseline$unemployment_rate_Med / df_baseline$unemployment_rate_Med)^2 -
    2 * df$corr_scenario * (df$unemployment_rate_Med / df$unemployment_rate_Med) * (df_baseline$unemployment_rate_Med / df_baseline$unemployment_rate_Med)
  )

  df$ratio_unemployment_rate_ci <- 1.96 * (df$ratio_unemployment_rate_sd / sqrt(50))

  return(df)
})

# Assign the updated data frames back to their respective variables
df_baseline <- scenarios_carbontax[[1]]
df_scenario2 <- scenarios_carbontax[[2]]
df_scenario3 <- scenarios_carbontax[[3]]
df_scenario4 <- scenarios_carbontax[[4]]
df_scenario5 <- scenarios_carbontax[[5]]
df_scenario6 <- scenarios_carbontax[[6]]

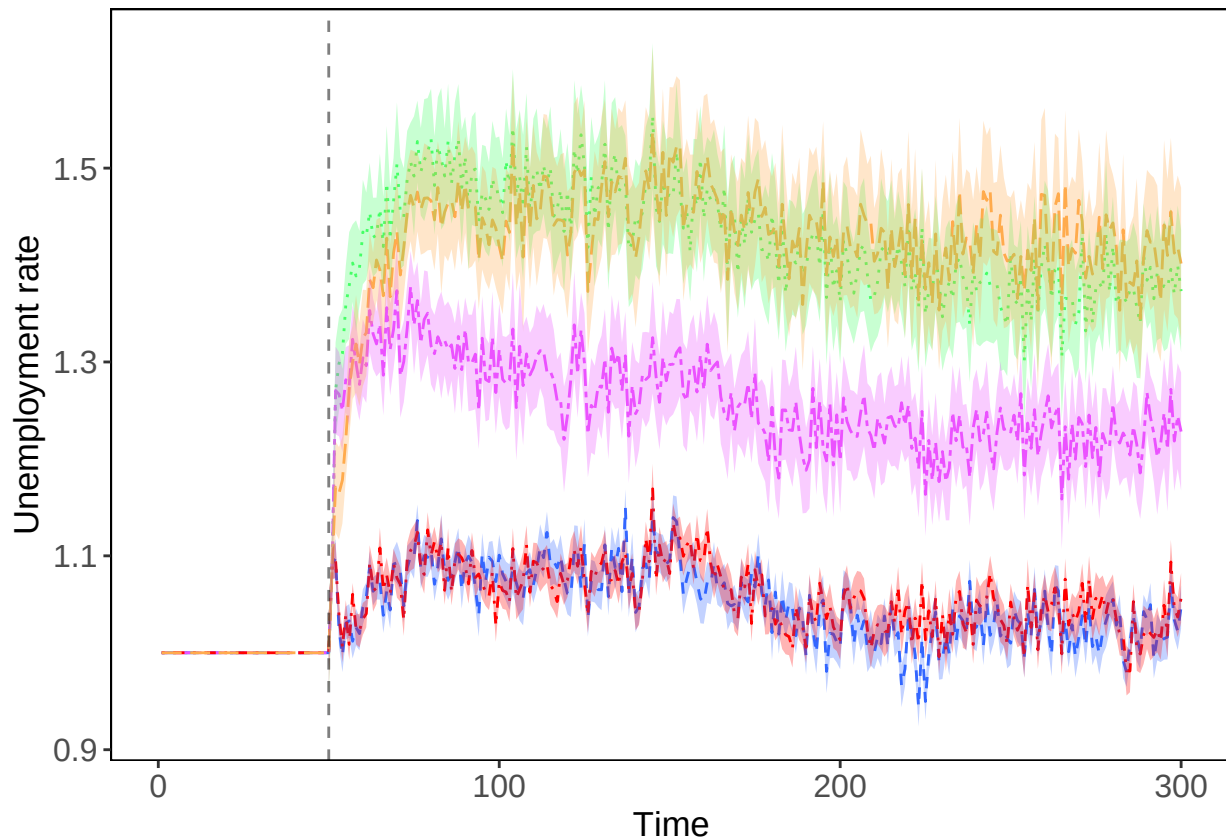
ggplot() +
```



```

geom_line(data = df_scenario2, aes(x = periods, y = ratio_unemployment_rate, color = "Carbon Tax"), l
geom_line(data = df_scenario3, aes(x = periods, y = ratio_unemployment_rate, color = "Carbon Tax and
geom_line(data = df_scenario4, aes(x = periods, y = ratio_unemployment_rate, color = "Carbon Tax and
geom_line(data = df_scenario5, aes(x = periods, y = ratio_unemployment_rate, color = "Carbon Tax and
geom_line(data = df_scenario6, aes(x = periods, y = ratio_unemployment_rate, color = "Carbon Tax and
geom_ribbon(data = subset(df_scenario2, periods >= 50), aes(x = periods, ymin = ratio_unemployment_rate,
geom_ribbon(data = subset(df_scenario3, periods >= 50), aes(x = periods, ymin = ratio_unemployment_rate,
geom_ribbon(data = subset(df_scenario4, periods >= 50), aes(x = periods, ymin = ratio_unemployment_rate,
geom_ribbon(data = subset(df_scenario5, periods >= 50), aes(x = periods, ymin = ratio_unemployment_rate,
geom_ribbon(data = subset(df_scenario6, periods >= 50), aes(x = periods, ymin = ratio_unemployment_rate,
theme(legend.position = "bottom") +
labs(x = "Time", y = "Unemployment rate", color = "", fill = "") +
theme(panel.border = element_rect(color = "black", size = 0.5, fill = NA),
      axis.title.x = element_text(size = 13),
      axis.title.y = element_text(size = 13),
      axis.text.x = element_text(size = 12),
      axis.text.y = element_text(size = 12),
      panel.background = element_rect(fill = "white"),
      legend.key = element_blank(),
      legend.position = "none",
      legend.background = element_blank()) +
scale_color_manual(values = c("#45ff66", "#EB52ff", "#3366ff", "#FFAB4D", "#ff0000"),
                   guide = guide_legend(override.aes = list(linetype = c("dotted", "dashed", "twodash
scale_fill_manual(values = c("#45ff66", "#EB52ff", "#3366ff", "#FFAB4D", "#ff0000")) +
geom_vline(xintercept = 50, linetype = "dashed", alpha = 0.5)

```



df_baseline\$inflation_Med

```
## [1] 0.015001867 0.018811861 0.016597633 0.018692862 0.017079960 0.019920280
## [7] 0.018192530 0.019073948 0.017734122 0.018135057 0.020734446 0.019891152
## [13] 0.016252310 0.021462994 0.016692053 0.017403037 0.018062723 0.019721491
## [19] 0.017499728 0.019080514 0.018644889 0.021771147 0.018508376 0.018354447
## [25] 0.019716795 0.015554655 0.018473759 0.016500319 0.019926236 0.018559803
## [31] 0.016082779 0.019596203 0.017394451 0.017253595 0.016742639 0.017378377
## [37] 0.018204647 0.015405653 0.019660233 0.016391172 0.016981843 0.015791045
## [43] 0.014383047 0.017222848 0.014853052 0.018051681 0.016336150 0.017784203
## [49] 0.014625741 0.019485244 0.012528260 0.016369447 0.014449350 0.016951332
## [55] 0.014635341 0.015834052 0.014186641 0.017237739 0.014594470 0.013653338
## [61] 0.012465820 0.016100947 0.016047262 0.011768893 0.017626301 0.012448582
## [67] 0.014829121 0.015459987 0.015013940 0.015272688 0.016191129 0.016785014
## [73] 0.013834827 0.012384698 0.014133439 0.016067318 0.011002864 0.014276123
## [79] 0.010496910 0.013783998 0.013198221 0.016807948 0.011026091 0.013101896
## [85] 0.014122178 0.009961259 0.010661017 0.015707611 0.014898004 0.011179009
## [91] 0.012584272 0.013095521 0.010374148 0.010283507 0.015647493 0.014917337
## [97] 0.011041323 0.015166693 0.012243761 0.012480661 0.009177433 0.011762044
## [103] 0.008719103 0.016525700 0.008218462 0.017029738 0.010547661 0.014604894
## [109] 0.007872920 0.012188686 0.010781615 0.008193735 0.015238109 0.009456683
## [115] 0.012649649 0.008119696 0.011198350 0.012322427 0.010890829 0.008977416
## [121] 0.009772760 0.012359884 0.012615110 0.012810513 0.011928590 0.007895613
## [127] 0.010376474 0.009170919 0.009351301 0.010242145 0.015768420 0.007915938
## [133] 0.007085791 0.014490460 0.010540861 0.009259805 0.007078357 0.010707588
## [139] 0.006989752 0.008662438 0.010989545 0.013029677 0.007996814 0.010995966
## [145] 0.011566869 0.013963925 0.010281209 0.008558944 0.011133793 0.011618130
## [151] 0.010801334 0.006599581 0.011362545 0.009026406 0.010562611 0.006987878
## [157] 0.009198914 0.015792499 0.010044229 0.014071491 0.008776467 0.010426823
## [163] 0.011017371 0.009724955 0.009433455 0.013296352 0.009957212 0.010219410
## [169] 0.007097429 0.010773225 0.009000438 0.007003142 0.007822935 0.009345044
## [175] 0.011797907 0.007323219 0.011009256 0.006339824 0.010954074 0.008883530
## [181] 0.005140887 0.012640278 0.009616062 0.008666100 0.012891539 0.007475128
## [187] 0.014031652 0.009015465 0.013050283 0.009290665 0.010624464 0.014345445
## [193] 0.002392872 0.007448934 0.009480962 0.005714031 0.012301223 0.005171992
## [199] 0.007730672 0.003542427 0.012820306 0.008084306 0.006942614 0.007440158
## [205] 0.005910990 0.013670155 0.013454692 0.010438212 0.010304014 0.011476021
## [211] 0.012064497 0.012076752 0.007875389 0.011563763 0.011712849 0.006567362
## [217] 0.008666736 0.009292115 0.008449291 0.013927926 0.007663553 0.013412944
## [223] 0.003722227 0.014950461 0.003592471 0.012113021 0.003131292 0.011151262
## [229] 0.008827717 0.006752888 0.006082778 0.008811448 0.011245238 0.010623113
## [235] 0.009695766 0.011572847 0.004536570 0.011025740 0.013388013 0.009395483
## [241] 0.007729231 0.009047096 0.008141836 0.006233319 0.011676078 0.007131084
## [247] 0.011092557 0.010357637 0.010655486 0.009920563 0.008527004 0.010442792
## [253] 0.012307042 0.004713204 0.014295227 0.006136908 0.012505623 0.007554826
## [259] 0.009914661 0.011533744 0.010397268 0.007769604 0.009021248 0.006866627
## [265] 0.009865100 0.007518748 0.003013689 0.012276457 0.008389743 0.007807704
## [271] 0.010463148 0.001059257 0.013469041 0.004610410 0.008674290 0.007546061
## [277] 0.007258903 0.009752362 0.008785240 0.016528424 0.008138663 0.010376356
## [283] 0.008971078 0.006269507 0.007980060 0.009871612 0.015615642 0.006928684
## [289] 0.011631967 0.005968373 0.010136442 0.010612417 0.009681830 0.010023520
## [295] 0.013115765 0.009713595 0.013277954 0.006413155 0.006687738 0.007252165
```

```

# Apply the calculations to each scenario
scenarios_carbontax <- lapply(scenarios_carbontax, function(df) {
  df$ratio_cpi <- df$cpi_Med / df_baseline$cpi_Med

  df$corr_scenario <- cor(df$cpi_Med, df_baseline$cpi_Med)

  df$ratio_cpi_sd <- abs(df$ratio_cpi) * sqrt(
    (df$cpi_Med / df$cpi_Med)^2 +
    (df_baseline$cpi_Med / df_baseline$cpi_Med)^2 -
    2 * df$corr_scenario * (df$cpi_Med / df$cpi_Med) * (df_baseline$cpi_Med / df_baseline$cpi_Med)
  )

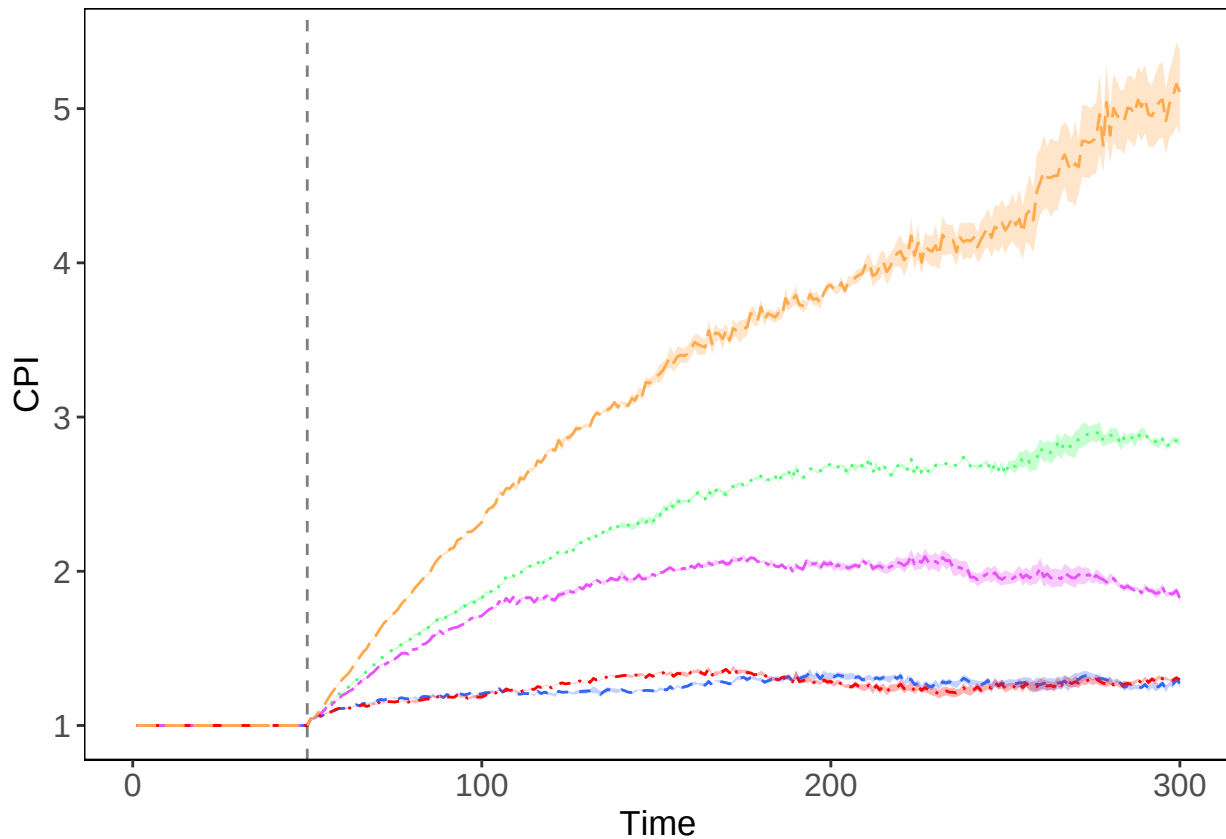
  df$ratio_cpi_ci <- 1.96 * (df$ratio_cpi_sd / sqrt(50))

  return(df)
})

# Assign the updated data frames back to their respective variables
df_baseline <- scenarios_carbontax[[1]]
df_scenario2 <- scenarios_carbontax[[2]]
df_scenario3 <- scenarios_carbontax[[3]]
df_scenario4 <- scenarios_carbontax[[4]]
df_scenario5 <- scenarios_carbontax[[5]]
df_scenario6 <- scenarios_carbontax[[6]]

ggplot() +
  geom_line(data = df_scenario2, aes(x = periods, y = ratio_cpi, color = "Carbon Tax"), linetype = "dotted") +
  geom_line(data = df_scenario3, aes(x = periods, y = ratio_cpi, color = "Carbon Tax and green public l"), linetype = "dotted") +
  geom_line(data = df_scenario4, aes(x = periods, y = ratio_cpi, color = "Carbon Tax and green price su"), linetype = "dotted") +
  geom_line(data = df_scenario5, aes(x = periods, y = ratio_cpi, color = "Carbon Tax and green research"), linetype = "dotted") +
  geom_line(data = df_scenario6, aes(x = periods, y = ratio_cpi, color = "Carbon Tax and green public p"), linetype = "dotted") +
  geom_ribbon(data = subset(df_scenario2, periods >= 50), aes(x = periods, ymin = ratio_cpi - ratio_cpi_ci, ymax = ratio_cpi + ratio_cpi_ci, color = "Carbon Tax"), linetype = "dotted") +
  geom_ribbon(data = subset(df_scenario3, periods >= 50), aes(x = periods, ymin = ratio_cpi - ratio_cpi_ci, ymax = ratio_cpi + ratio_cpi_ci, color = "Carbon Tax and green public l"), linetype = "dotted") +
  geom_ribbon(data = subset(df_scenario4, periods >= 50), aes(x = periods, ymin = ratio_cpi - ratio_cpi_ci, ymax = ratio_cpi + ratio_cpi_ci, color = "Carbon Tax and green price su"), linetype = "dotted") +
  geom_ribbon(data = subset(df_scenario5, periods >= 50), aes(x = periods, ymin = ratio_cpi - ratio_cpi_ci, ymax = ratio_cpi + ratio_cpi_ci, color = "Carbon Tax and green research"), linetype = "dotted") +
  geom_ribbon(data = subset(df_scenario6, periods >= 50), aes(x = periods, ymin = ratio_cpi - ratio_cpi_ci, ymax = ratio_cpi + ratio_cpi_ci, color = "Carbon Tax and green public p"), linetype = "dotted") +
  theme(legend.position = "bottom") +
  labs(x = "Time", y = "CPI", color = "", fill = "") +
  theme(panel.border = element_rect(color = "black", size = 0.5, fill = NA),
        axis.title.x = element_text(size = 13),
        axis.title.y = element_text(size = 13),
        axis.text.x = element_text(size = 12),
        axis.text.y = element_text(size = 12),
        panel.background = element_rect(fill = "white"),
        legend.key = element_blank(),
        legend.position = "none",
        legend.background = element_blank()) +
  scale_color_manual(values = c("#45ff66", "#EB52ff", "#3366ff", "#FFAB4D", "#ff0000"),
                     guide = guide_legend(override.aes = list(linetype = c("dotted", "dashed", "twodash"), fill = "white"))) +
  scale_fill_manual(values = c("#45ff66", "#EB52ff", "#3366ff", "#FFAB4D", "#ff0000")) +
  geom_vline(xintercept = 50, linetype = "dashed", alpha = 0.5)

```



```
# Apply the calculations to each scenario
scenarios_wealthtax <- lapply(scenarios_wealthtax, function(df) {
  df$ratio_emissions_coefficient <- df$emissions_coefficient_Med / df$baseline$emissions_coefficient_Med

  df$corr_scenario <- cor(df$emissions_coefficient_Med, df$baseline$emissions_coefficient_Med)

  df$ratio_emissions_coefficient_sd <- abs(df$ratio_emissions_coefficient) * sqrt(
    (df$emissions_coefficient_Mad / df$emissions_coefficient_Med)^2 +
    (df$baseline$emissions_coefficient_Mad / df$baseline$emissions_coefficient_Med)^2 -
    2 * df$corr_scenario * (df$emissions_coefficient_Mad / df$emissions_coefficient_Med) * (df$baseline$emissions_coefficient_Mad / df$baseline$emissions_coefficient_Med)
  )

  df$ratio_emissions_coefficient_ci <- 1.96 * (df$ratio_emissions_coefficient_sd / sqrt(50))

  return(df)
})

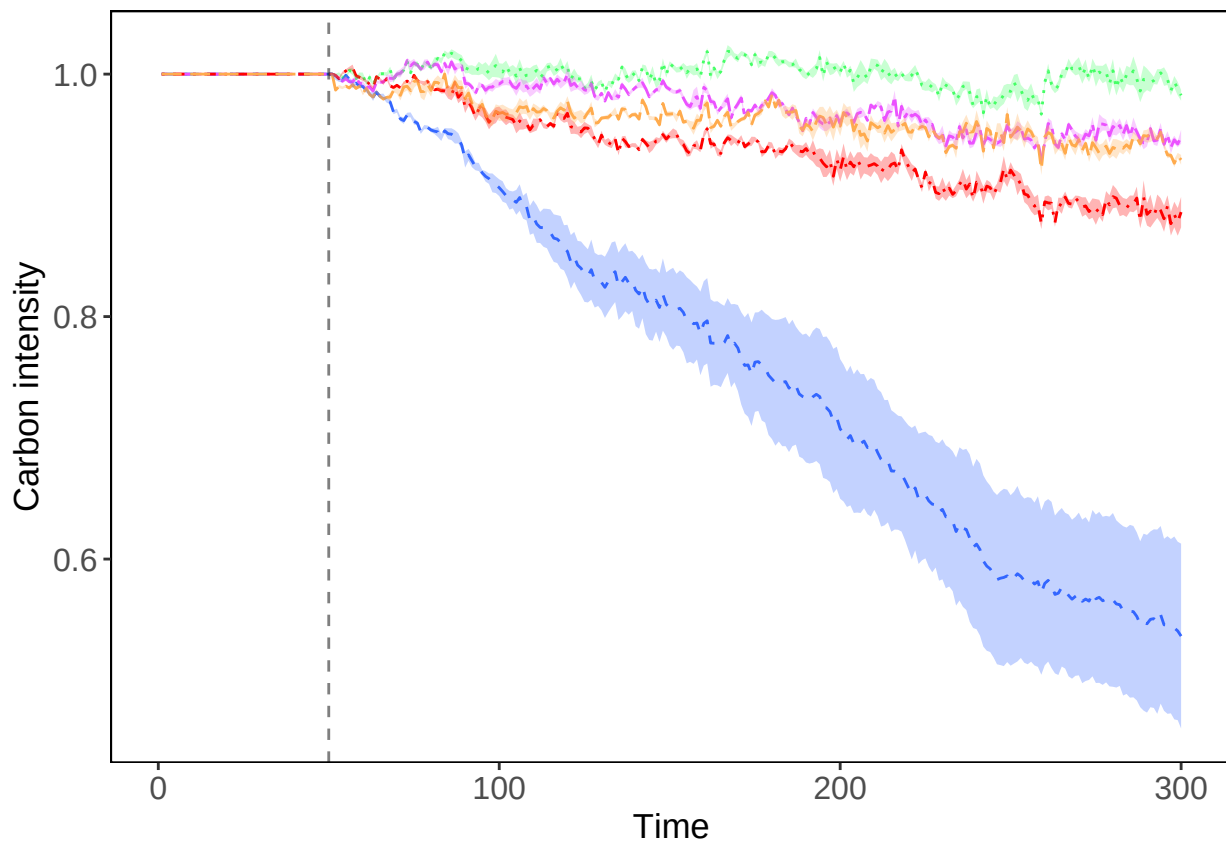
# Assign the updated data frames back to their respective variables
df_baseline <- scenarios_wealthtax[[1]]
df_scenario7 <- scenarios_wealthtax[[2]]
df_scenario8 <- scenarios_wealthtax[[3]]
df_scenario9 <- scenarios_wealthtax[[4]]
df_scenario10 <- scenarios_wealthtax[[5]]
df_scenario11 <- scenarios_wealthtax[[6]]

ggplot() +
```

```

geom_line(data = df_scenario7, aes(x = periods, y = ratio_emissions_coefficient, color = "Wealth Tax"),
geom_line(data = df_scenario8, aes(x = periods, y = ratio_emissions_coefficient, color = "Wealth Tax"),
geom_line(data = df_scenario9, aes(x = periods, y = ratio_emissions_coefficient, color = "Wealth Tax"),
geom_line(data = df_scenario10, aes(x = periods, y = ratio_emissions_coefficient, color = "Wealth Tax"),
geom_line(data = df_scenario11, aes(x = periods, y = ratio_emissions_coefficient, color = "Wealth Tax"),
geom_ribbon(data = subset(df_scenario7, periods >= 50), aes(x = periods, ymin = ratio_emissions_coefficient, ymax = ratio_emissions_coefficient),
geom_ribbon(data = subset(df_scenario8, periods >= 50), aes(x = periods, ymin = ratio_emissions_coefficient, ymax = ratio_emissions_coefficient),
geom_ribbon(data = subset(df_scenario9, periods >= 50), aes(x = periods, ymin = ratio_emissions_coefficient, ymax = ratio_emissions_coefficient),
geom_ribbon(data = subset(df_scenario10, periods >= 50), aes(x = periods, ymin = ratio_emissions_coefficient, ymax = ratio_emissions_coefficient),
geom_ribbon(data = subset(df_scenario11, periods >= 50), aes(x = periods, ymin = ratio_emissions_coefficient, ymax = ratio_emissions_coefficient),
theme(legend.position = "bottom") +
labs(x = "Time", y = "Carbon intensity", color = "", fill = "") +
theme(panel.border = element_rect(color = "black", size = 0.5, fill = NA),
axis.title.x = element_text(size = 13),
axis.title.y = element_text(size = 13),
axis.text.x = element_text(size = 12),
axis.text.y = element_text(size = 12),
panel.background = element_rect(fill = "white"),
legend.key = element_blank(),
legend.position = "none",
legend.background = element_blank()) +
scale_color_manual(values = c("#45ff66", "#EB52ff", "#3366ff", "#FFAB4D", "#ff0000"),
guide = guide_legend(override.aes = list(linetype = c("dotted", "dashed", "twodash", "solid", "longdash"),
scale_fill_manual(values = c("#45ff66", "#EB52ff", "#3366ff", "#FFAB4D", "#ff0000")) +
geom_vline(xintercept = 50, linetype = "dashed", alpha = 0.5)

```



```

# Apply the calculations to each scenario
scenarios_wealthtax <- lapply(scenarios_wealthtax, function(df) {
  df$ratio_gini_hh <- df$gini_hh_Med / df_baseline$gini_hh_Med

  df$corr_scenario <- cor(df$gini_hh_Med, df_baseline$gini_hh_Med)

  df$ratio_gini_hh_sd <- abs(df$ratio_gini_hh) * sqrt(
    (df$gini_hh_Mad / df$gini_hh_Med)^2 +
    (df_baseline$gini_hh_Mad / df_baseline$gini_hh_Med)^2 -
    2 * df$corr_scenario * (df$gini_hh_Mad / df$gini_hh_Med) * (df_baseline$gini_hh_Mad / df_baseline$gini_hh_Med)
  )

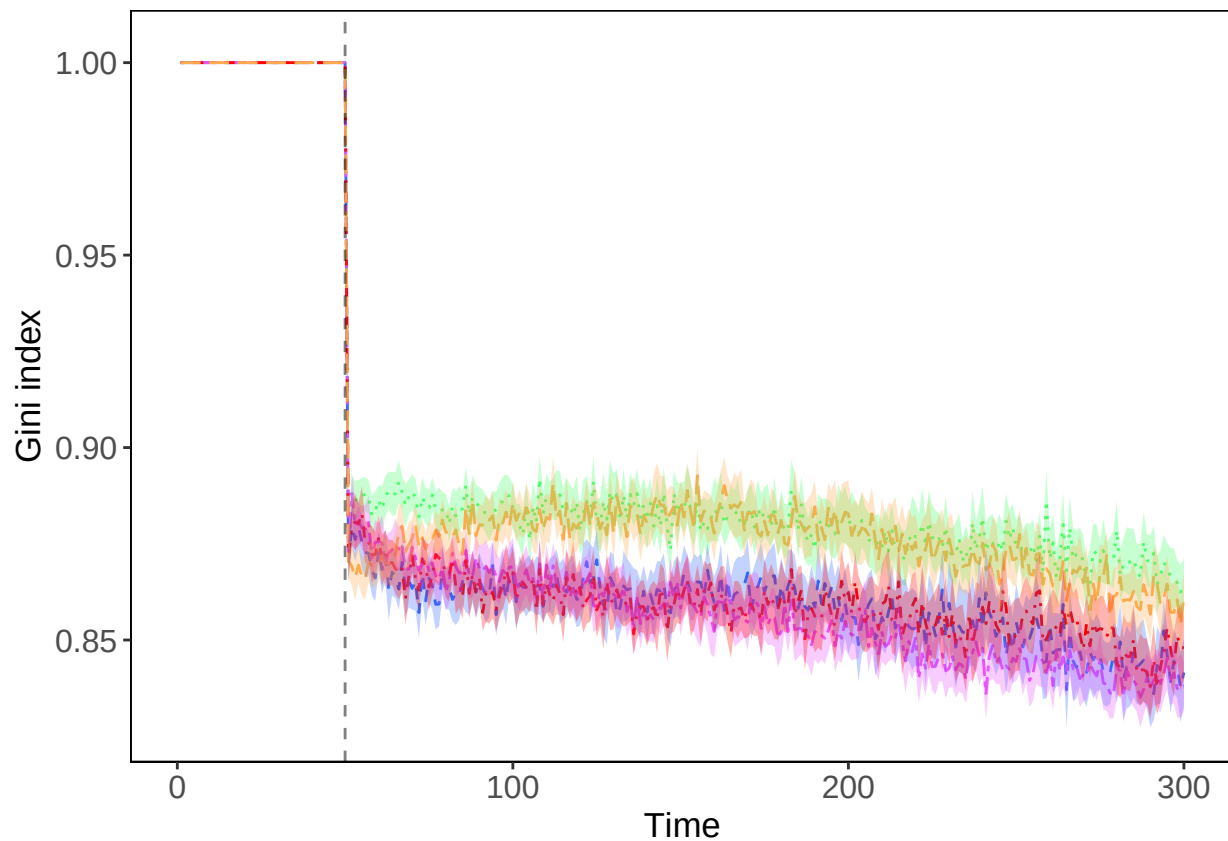
  df$ratio_gini_hh_ci <- 1.96 * (df$ratio_gini_hh_sd / sqrt(50))

  return(df)
})

# Assign the updated data frames back to their respective variables
df_baseline <- scenarios_wealthtax[[1]]
df_scenario7 <- scenarios_wealthtax[[2]]
df_scenario8 <- scenarios_wealthtax[[3]]
df_scenario9 <- scenarios_wealthtax[[4]]
df_scenario10 <- scenarios_wealthtax[[5]]
df_scenario11 <- scenarios_wealthtax[[6]]

ggplot() +
  geom_line(data = df_scenario7, aes(x = periods, y = ratio_gini_hh, color = "Wealth Tax"), linetype = "dotted") +
  geom_line(data = df_scenario8, aes(x = periods, y = ratio_gini_hh, color = "Wealth Tax and green public goods"), linetype = "dashed") +
  geom_line(data = df_scenario9, aes(x = periods, y = ratio_gini_hh, color = "Wealth Tax and green price index"), linetype = "twodash") +
  geom_line(data = df_scenario10, aes(x = periods, y = ratio_gini_hh, color = "Wealth Tax and green residential property"), linetype = "dotted") +
  geom_line(data = df_scenario11, aes(x = periods, y = ratio_gini_hh, color = "Wealth Tax and green public goods"), linetype = "dashed") +
  geom_ribbon(data = subset(df_scenario7, periods >= 50), aes(x = periods, ymin = ratio_gini_hh - ratio_gini_hh_ci, ymax = ratio_gini_hh + ratio_gini_hh_ci, color = "Wealth Tax"), fill = "#45ff66", alpha = 0.5) +
  geom_ribbon(data = subset(df_scenario8, periods >= 50), aes(x = periods, ymin = ratio_gini_hh - ratio_gini_hh_ci, ymax = ratio_gini_hh + ratio_gini_hh_ci, color = "Wealth Tax and green public goods"), fill = "#EB52ff", alpha = 0.5) +
  geom_ribbon(data = subset(df_scenario9, periods >= 50), aes(x = periods, ymin = ratio_gini_hh - ratio_gini_hh_ci, ymax = ratio_gini_hh + ratio_gini_hh_ci, color = "Wealth Tax and green price index"), fill = "#3366ff", alpha = 0.5) +
  geom_ribbon(data = subset(df_scenario10, periods >= 50), aes(x = periods, ymin = ratio_gini_hh - ratio_gini_hh_ci, ymax = ratio_gini_hh + ratio_gini_hh_ci, color = "Wealth Tax and green residential property"), fill = "#FFAB4D", alpha = 0.5) +
  geom_ribbon(data = subset(df_scenario11, periods >= 50), aes(x = periods, ymin = ratio_gini_hh - ratio_gini_hh_ci, ymax = ratio_gini_hh + ratio_gini_hh_ci, color = "Wealth Tax and green public goods"), fill = "#ff0000", alpha = 0.5) +
  theme(legend.position = "bottom") +
  labs(x = "Time", y = "Gini index", color = "", fill = "") +
  theme(panel.border = element_rect(color = "black", size = 0.5, fill = NA),
    axis.title.x = element_text(size = 13),
    axis.title.y = element_text(size = 13),
    axis.text.x = element_text(size = 12),
    axis.text.y = element_text(size = 12),
    panel.background = element_rect(fill = "white"),
    legend.key = element_blank(),
    legend.position = "none",
    legend.background = element_blank()) +
  scale_color_manual(values = c("#45ff66", "#EB52ff", "#3366ff", "#FFAB4D", "#ff0000"),
    guide = guide_legend(override.aes = list(linetype = c("dotted", "dashed", "twodash")))) +
  scale_fill_manual(values = c("#45ff66", "#EB52ff", "#3366ff", "#FFAB4D", "#ff0000")) +
  geom_vline(xintercept = 50, linetype = "dashed", alpha = 0.5)

```



```
# Apply the calculations to each scenario
scenarios_wealthtax <- lapply(scenarios_wealthtax, function(df) {
  df$ratio_wage_share <- df$wage_share_Med / df_baseline$wage_share_Med

  df$corr_scenario <- cor(df$wage_share_Med, df_baseline$wage_share_Med)

  df$ratio_wage_share_sd <- abs(df$ratio_wage_share) * sqrt(
    (df$wage_share_Mad / df$wage_share_Med)^2 +
    (df_baseline$wage_share_Mad / df_baseline$wage_share_Med)^2 -
    2 * df$corr_scenario * (df$wage_share_Mad / df$wage_share_Med) * (df_baseline$wage_share_Mad / df_b
  )

  df$ratio_wage_share_ci <- 1.96 * (df$ratio_wage_share_sd / sqrt(50))

  return(df)
})

# Assign the updated data frames back to their respective variables
df_baseline <- scenarios_wealthtax[[1]]
df_scenario7 <- scenarios_wealthtax[[2]]
df_scenario8 <- scenarios_wealthtax[[3]]
df_scenario9 <- scenarios_wealthtax[[4]]
df_scenario10 <- scenarios_wealthtax[[5]]
df_scenario11 <- scenarios_wealthtax[[6]]

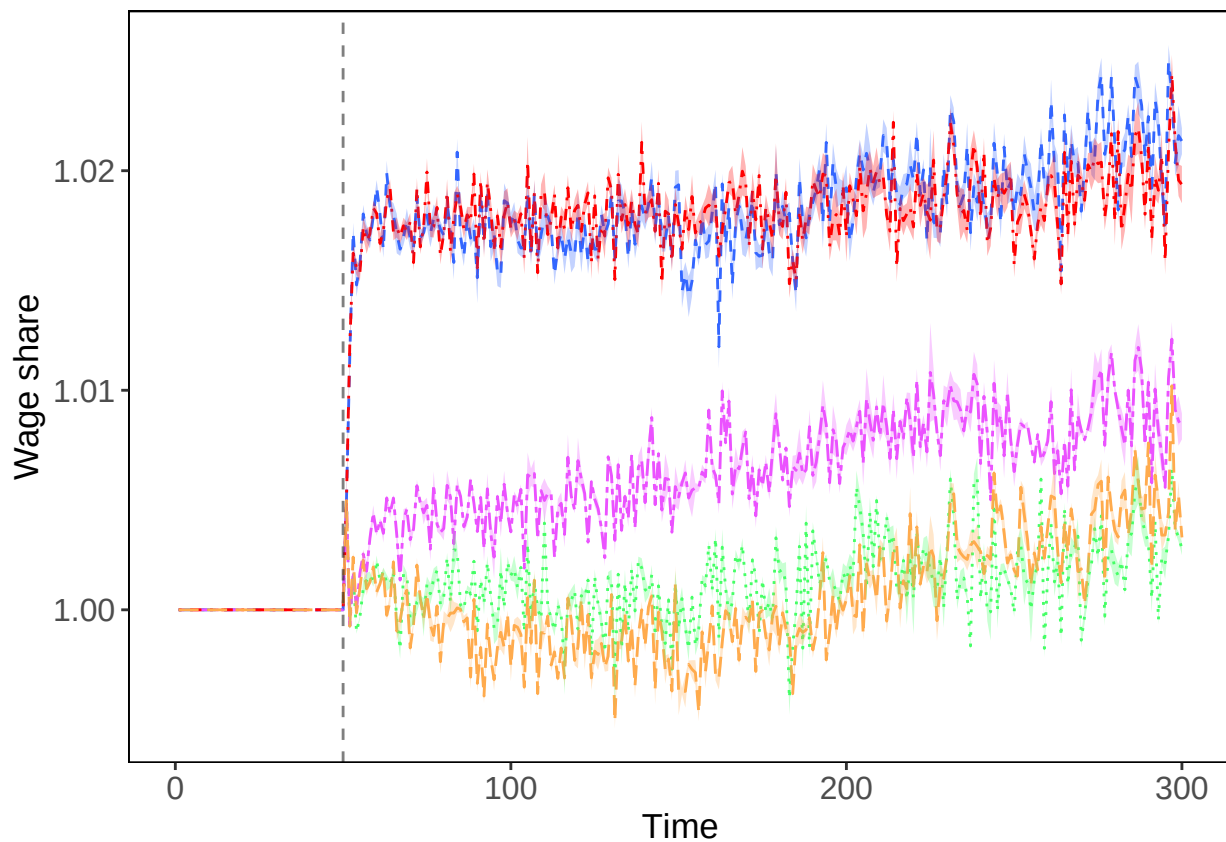
ggplot() +
```



```

geom_line(data = df_scenario7, aes(x = periods, y = ratio_wage_share, color = "Wealth Tax"), linetype
geom_line(data = df_scenario8, aes(x = periods, y = ratio_wage_share, color = "Wealth Tax and green p
geom_line(data = df_scenario9, aes(x = periods, y = ratio_wage_share, color = "Wealth Tax and green p
geom_line(data = df_scenario10, aes(x = periods, y = ratio_wage_share, color = "Wealth Tax and green
geom_line(data = df_scenario11, aes(x = periods, y = ratio_wage_share, color = "Wealth Tax and green
geom_ribbon(data = subset(df_scenario7, periods >=50), aes(x = periods, ymin = ratio_wage_share - rat
geom_ribbon(data = subset(df_scenario8, periods >=50), aes(x = periods, ymin = ratio_wage_share - rat
geom_ribbon(data = subset(df_scenario9, periods >=50), aes(x = periods, ymin = ratio_wage_share - rat
geom_ribbon(data = subset(df_scenario10, periods >=50), aes(x = periods, ymin = ratio_wage_share - ra
geom_ribbon(data = subset(df_scenario11, periods >=50), aes(x = periods, ymin = ratio_wage_share - ra
theme(legend.position = "bottom") +
labs(x = "Time", y = "Wage share", color = "", fill = "") +
theme(panel.border = element_rect(color = "black", size = 0.5, fill = NA),
      axis.title.x = element_text(size = 13),
      axis.title.y = element_text(size = 13),
      axis.text.x = element_text(size = 12),
      axis.text.y = element_text(size = 12),
      panel.background = element_rect(fill = "white"),
      legend.key = element_blank(),
      legend.position = "none",
      legend.background = element_blank()) +
scale_color_manual(values = c("#45ff66", "#EB52ff", "#3366ff", "#FFAB4D", "#ff0000"),
                   guide = guide_legend(override.aes = list(linetype = c("dotted", "dashed", "twodash
scale_fill_manual(values = c("#45ff66", "#EB52ff", "#3366ff", "#FFAB4D", "#ff0000")) +
geom_vline(xintercept = 50, linetype = "dashed", alpha = 0.5)

```




```

# Apply the calculations to each scenario
scenarios_wealthtax <- lapply(scenarios_wealthtax, function(df) {
  df$ratio_gdp_real <- df$gdp_real_Med / df_baseline$gdp_real_Med

  df$corr_scenario <- cor(df$gdp_real_Med, df_baseline$gdp_real_Med)

  df$ratio_gdp_real_sd <- abs(df$ratio_gdp_real) * sqrt(
    (df$gdp_real_Med / df$gdp_real_Med)^2 +
    (df_baseline$gdp_real_Med / df_baseline$gdp_real_Med)^2 -
    2 * df$corr_scenario * (df$gdp_real_Med / df$gdp_real_Med) * (df_baseline$gdp_real_Med / df_baseline$gdp_real_Med)
  )

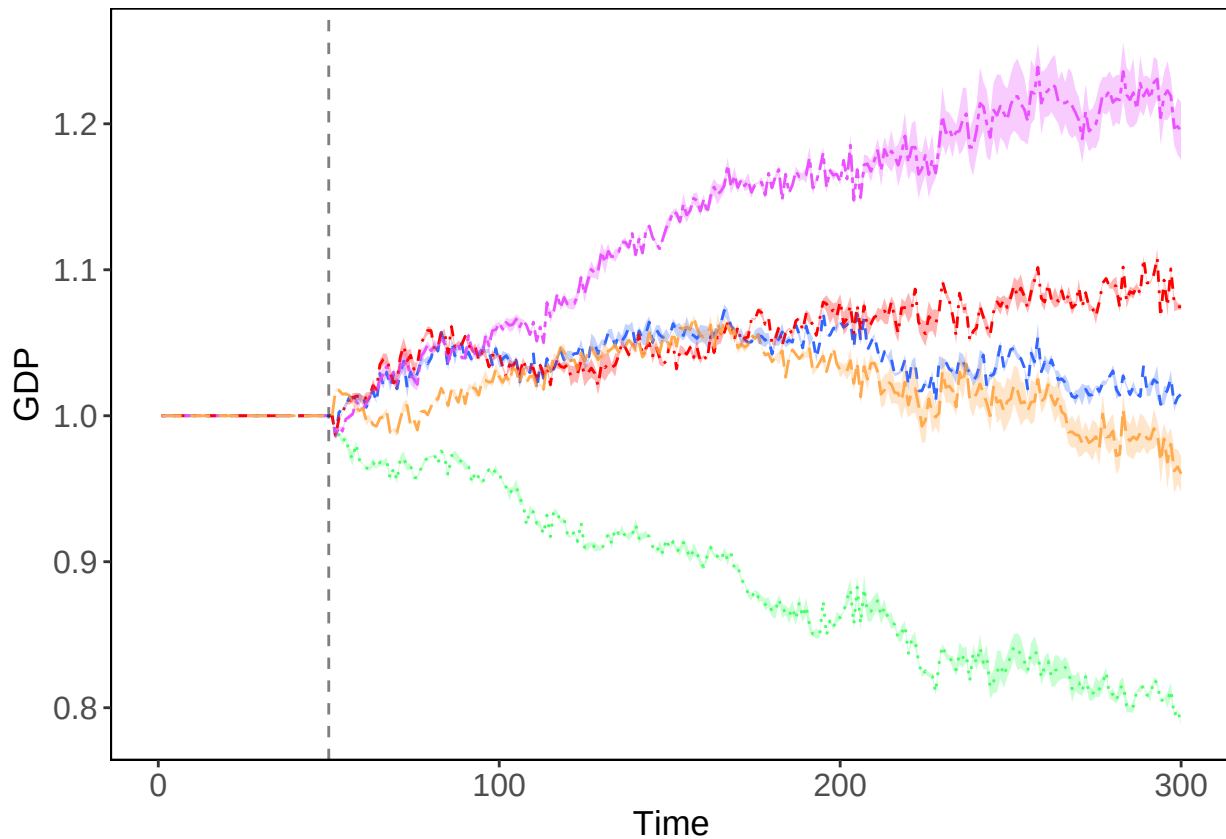
  df$ratio_gdp_real_ci <- 1.96 * (df$ratio_gdp_real_sd / sqrt(50))

  return(df)
})

# Assign the updated data frames back to their respective variables
df_baseline <- scenarios_wealthtax[[1]]
df_scenario7 <- scenarios_wealthtax[[2]]
df_scenario8 <- scenarios_wealthtax[[3]]
df_scenario9 <- scenarios_wealthtax[[4]]
df_scenario10 <- scenarios_wealthtax[[5]]
df_scenario11 <- scenarios_wealthtax[[6]]

ggplot() +
  geom_line(data = df_scenario7, aes(x = periods, y = ratio_gdp_real, color = "Wealth Tax"), linetype = "dotted") +
  geom_line(data = df_scenario8, aes(x = periods, y = ratio_gdp_real, color = "Wealth Tax and green public goods"), linetype = "dashed") +
  geom_line(data = df_scenario9, aes(x = periods, y = ratio_gdp_real, color = "Wealth Tax and green public goods and green public goods"), linetype = "twodash") +
  geom_line(data = df_scenario10, aes(x = periods, y = ratio_gdp_real, color = "Wealth Tax and green public goods and green public goods and green public goods"), linetype = "solid") +
  geom_line(data = df_scenario11, aes(x = periods, y = ratio_gdp_real, color = "Wealth Tax and green public goods and green public goods and green public goods and green public goods"), linetype = "solid") +
  geom_ribbon(data = subset(df_scenario7, periods >= 50), aes(x = periods, ymin = ratio_gdp_real - ratio_gdp_real_ci, ymax = ratio_gdp_real + ratio_gdp_real_ci, color = "Wealth Tax"), fill = "#45ff66", alpha = 0.5) +
  geom_ribbon(data = subset(df_scenario8, periods >= 50), aes(x = periods, ymin = ratio_gdp_real - ratio_gdp_real_ci, ymax = ratio_gdp_real + ratio_gdp_real_ci, color = "Wealth Tax and green public goods"), fill = "#EB52ff", alpha = 0.5) +
  geom_ribbon(data = subset(df_scenario9, periods >= 50), aes(x = periods, ymin = ratio_gdp_real - ratio_gdp_real_ci, ymax = ratio_gdp_real + ratio_gdp_real_ci, color = "Wealth Tax and green public goods and green public goods"), fill = "#3366ff", alpha = 0.5) +
  geom_ribbon(data = subset(df_scenario10, periods >= 50), aes(x = periods, ymin = ratio_gdp_real - ratio_gdp_real_ci, ymax = ratio_gdp_real + ratio_gdp_real_ci, color = "Wealth Tax and green public goods and green public goods and green public goods"), fill = "#FFAB4D", alpha = 0.5) +
  geom_ribbon(data = subset(df_scenario11, periods >= 50), aes(x = periods, ymin = ratio_gdp_real - ratio_gdp_real_ci, ymax = ratio_gdp_real + ratio_gdp_real_ci, color = "Wealth Tax and green public goods and green public goods and green public goods and green public goods"), fill = "#ff0000", alpha = 0.5) +
  theme(legend.position = "bottom") +
  labs(x = "Time", y = "GDP", color = "", fill = "") +
  theme(panel.border = element_rect(color = "black", size = 0.5, fill = NA),
        axis.title.x = element_text(size = 13),
        axis.title.y = element_text(size = 13),
        axis.text.x = element_text(size = 12),
        axis.text.y = element_text(size = 12),
        panel.background = element_rect(fill = "white"),
        legend.key = element_blank(),
        legend.position = "none",
        legend.background = element_blank()) +
  scale_color_manual(values = c("#45ff66", "#EB52ff", "#3366ff", "#FFAB4D", "#ff0000"),
                     guide = guide_legend(override.aes = list(linetype = c("dotted", "dashed", "twodash", "solid", "solid")))) +
  scale_fill_manual(values = c("#45ff66", "#EB52ff", "#3366ff", "#FFAB4D", "#ff0000")) +
  geom_vline(xintercept = 50, linetype = "dashed", alpha = 0.5)

```



```
# Apply the calculations to each scenario
scenarios_wealthtax <- lapply(scenarios_wealthtax, function(df) {
  df$ratio_emissions <- df$emissions_Med / df_baseline$emissions_Med

  df$corr_scenario <- cor(df$emissions_Med, df_baseline$emissions_Med)

  df$ratio_emissions_sd <- abs(df$ratio_emissions) * sqrt(
    (df$emissions_Mad / df$emissions_Med)^2 +
    (df_baseline$emissions_Mad / df_baseline$emissions_Med)^2 -
    2 * df$corr_scenario * (df$emissions_Mad / df$emissions_Med) * (df_baseline$emissions_Mad / df_base
  )

  df$ratio_emissions_ci <- 1.96 * (df$ratio_emissions_sd / sqrt(50))

  return(df)
})

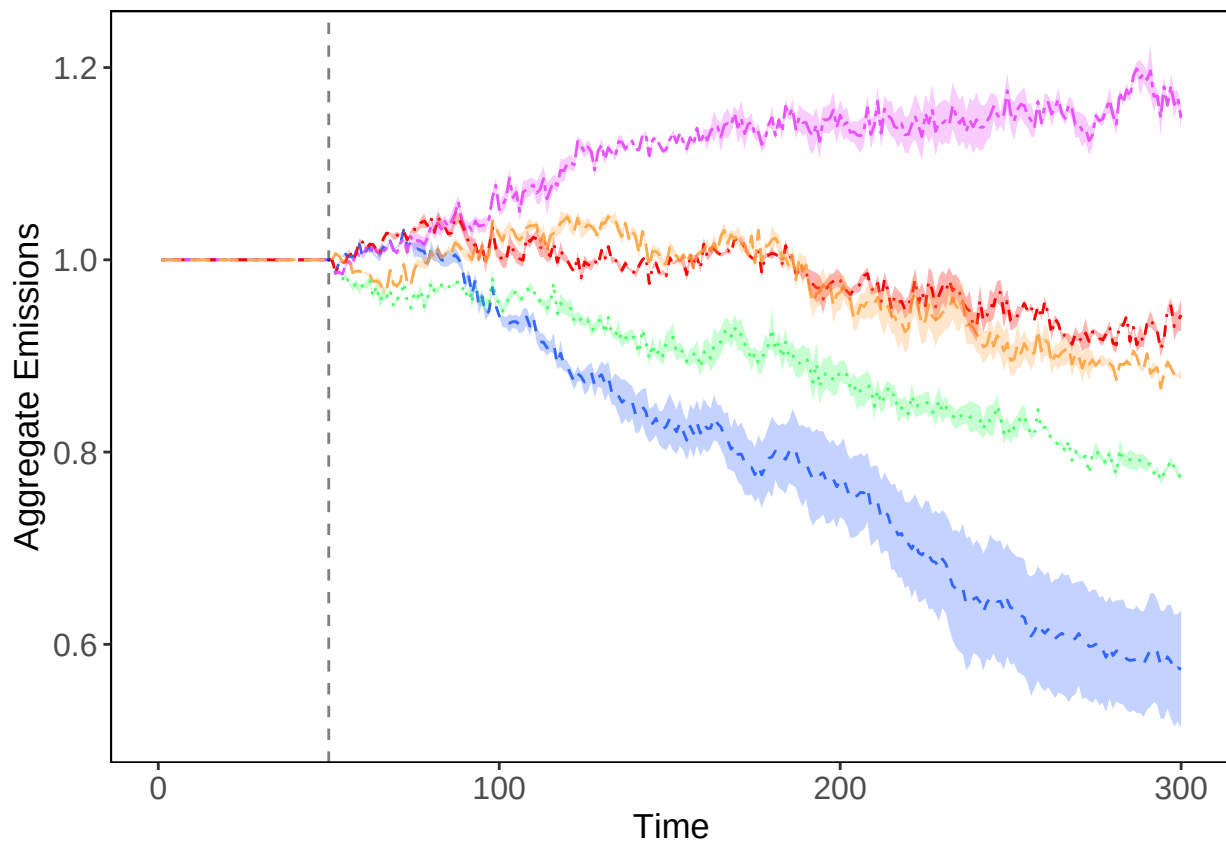
# Assign the updated data frames back to their respective variables
df_baseline <- scenarios_wealthtax[[1]]
df_scenario7 <- scenarios_wealthtax[[2]]
df_scenario8 <- scenarios_wealthtax[[3]]
df_scenario9 <- scenarios_wealthtax[[4]]
df_scenario10 <- scenarios_wealthtax[[5]]
df_scenario11 <- scenarios_wealthtax[[6]]

ggplot() +
```

```

geom_line(data = df_scenario7, aes(x = periods, y = ratio_emissions, color = "Wealth Tax"), linetype = "dotted"),
geom_line(data = df_scenario8, aes(x = periods, y = ratio_emissions, color = "Wealth Tax and green pul"), linetype = "dotted"),
geom_line(data = df_scenario9, aes(x = periods, y = ratio_emissions, color = "Wealth Tax and green pr"), linetype = "dotted"),
geom_line(data = df_scenario10, aes(x = periods, y = ratio_emissions, color = "Wealth Tax and green r"), linetype = "dotted"),
geom_line(data = df_scenario11, aes(x = periods, y = ratio_emissions, color = "Wealth Tax and green p"), linetype = "dotted"),
geom_ribbon(data = subset(df_scenario7, periods >= 50), aes(x = periods, ymin = ratio_emissions - ratio_emissions, ymax = ratio_emissions + ratio_emissions, color = "Wealth Tax"), linetype = "dotted"),
geom_ribbon(data = subset(df_scenario8, periods >= 50), aes(x = periods, ymin = ratio_emissions - ratio_emissions, ymax = ratio_emissions + ratio_emissions, color = "Wealth Tax and green pul"), linetype = "dotted"),
geom_ribbon(data = subset(df_scenario9, periods >= 50), aes(x = periods, ymin = ratio_emissions - ratio_emissions, ymax = ratio_emissions + ratio_emissions, color = "Wealth Tax and green pr"), linetype = "dotted"),
geom_ribbon(data = subset(df_scenario10, periods >= 50), aes(x = periods, ymin = ratio_emissions - ratio_emissions, ymax = ratio_emissions + ratio_emissions, color = "Wealth Tax and green r"), linetype = "dotted"),
geom_ribbon(data = subset(df_scenario11, periods >= 50), aes(x = periods, ymin = ratio_emissions - ratio_emissions, ymax = ratio_emissions + ratio_emissions, color = "Wealth Tax and green p"), linetype = "dotted"),
theme(legend.position = "bottom") +
labs(x = "Time", y = "Aggregate Emissions", color = "", fill = "") +
theme(panel.border = element_rect(color = "black", size = 0.5, fill = NA),
      axis.title.x = element_text(size = 13),
      axis.title.y = element_text(size = 13),
      axis.text.x = element_text(size = 12),
      axis.text.y = element_text(size = 12),
      panel.background = element_rect(fill = "white"),
      legend.key = element_blank(),
      legend.position = "none",
      legend.background = element_blank()) +
scale_color_manual(values = c("#45ff66", "#EB52ff", "#3366ff", "#FFAB4D", "#ff0000"),
                  guide = guide_legend(override.aes = list(linetype = c("dotted", "dashed", "twodash", "solid", "longdash"),
                  scale_fill_manual(values = c("#45ff66", "#EB52ff", "#3366ff", "#FFAB4D", "#ff0000")) +
geom_vline(xintercept = 50, linetype = "dashed", alpha = 0.5)

```



```

# Apply the calculations to each scenario
scenarios_wealthtax <- lapply(scenarios_wealthtax, function(df) {
  df$ratio_unemployment_rate <- df$unemployment_rate_Med / df_baseline$unemployment_rate_Med

  df$corr_scenario <- cor(df$unemployment_rate_Med, df_baseline$unemployment_rate_Med)

  df$ratio_unemployment_rate_sd <- abs(df$ratio_unemployment_rate) * sqrt(
    (df$unemployment_rate_Med / df$unemployment_rate_Med)^2 +
    (df_baseline$unemployment_rate_Med / df_baseline$unemployment_rate_Med)^2 -
    2 * df$corr_scenario * (df$unemployment_rate_Med / df$unemployment_rate_Med) * (df_baseline$unemployment_rate_Med / df_baseline$unemployment_rate_Med)
  )

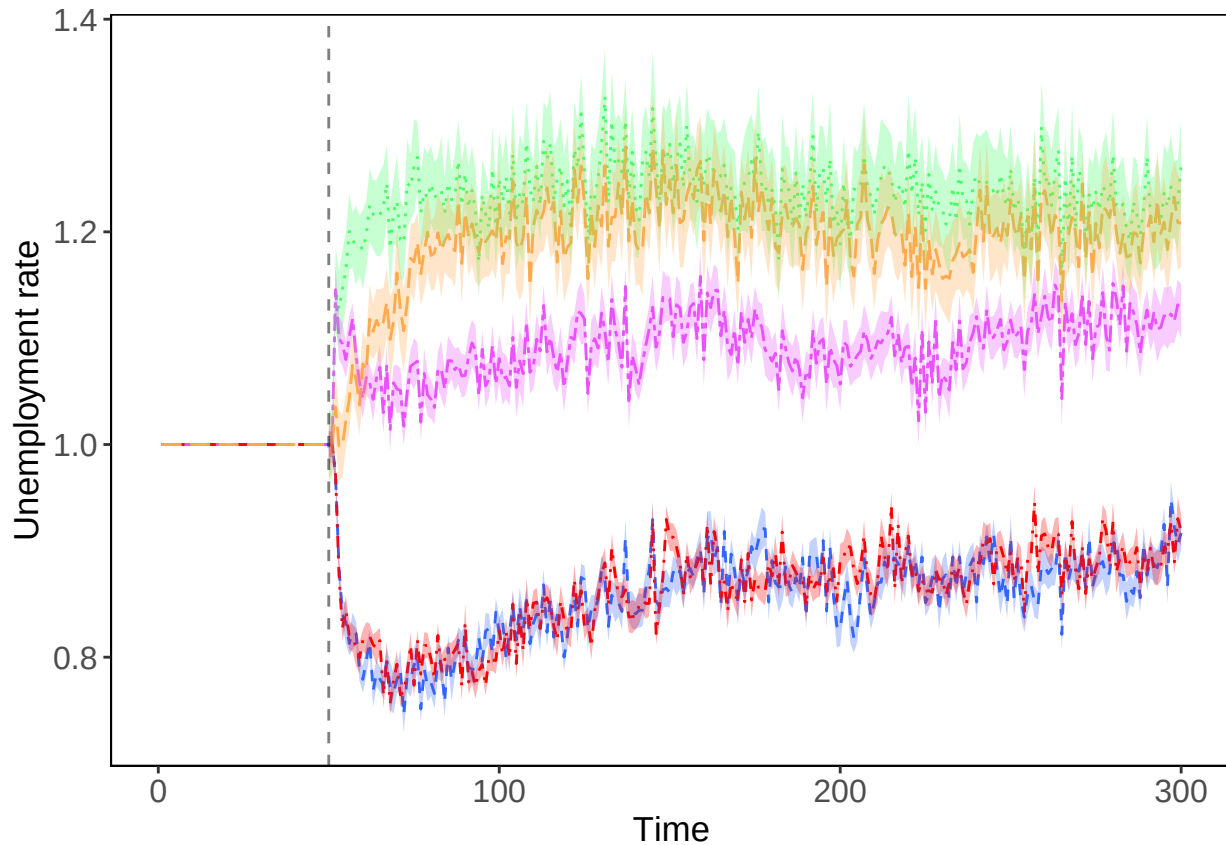
  df$ratio_unemployment_rate_ci <- 1.96 * (df$ratio_unemployment_rate_sd / sqrt(50))

  return(df)
})

# Assign the updated data frames back to their respective variables
df_baseline <- scenarios_wealthtax[[1]]
df_scenario7 <- scenarios_wealthtax[[2]]
df_scenario8 <- scenarios_wealthtax[[3]]
df_scenario9 <- scenarios_wealthtax[[4]]
df_scenario10 <- scenarios_wealthtax[[5]]
df_scenario11 <- scenarios_wealthtax[[6]]

ggplot() +
  geom_line(data = df_scenario7, aes(x = periods, y = ratio_unemployment_rate, color = "Wealth Tax"), lty = "dotted") +
  geom_line(data = df_scenario8, aes(x = periods, y = ratio_unemployment_rate, color = "Wealth Tax and g", lty = "dashed") +
  geom_line(data = df_scenario9, aes(x = periods, y = ratio_unemployment_rate, color = "Wealth Tax and g", lty = "twodash") +
  geom_line(data = df_scenario10, aes(x = periods, y = ratio_unemployment_rate, color = "Wealth Tax and g", lty = "dotted") +
  geom_line(data = df_scenario11, aes(x = periods, y = ratio_unemployment_rate, color = "Wealth Tax and g", lty = "dashed") +
  geom_ribbon(data = subset(df_scenario7, periods >= 50), aes(x = periods, ymin = ratio_unemployment_rate, ymax = ratio_unemployment_rate + 1.96 * ratio_unemployment_rate_sd / sqrt(50), color = "Wealth Tax and g", fill = "#45ff66", lty = "dotted") +
  geom_ribbon(data = subset(df_scenario8, periods >= 50), aes(x = periods, ymin = ratio_unemployment_rate, ymax = ratio_unemployment_rate + 1.96 * ratio_unemployment_rate_sd / sqrt(50), color = "Wealth Tax and g", fill = "#EB52ff", lty = "dashed") +
  geom_ribbon(data = subset(df_scenario9, periods >= 50), aes(x = periods, ymin = ratio_unemployment_rate, ymax = ratio_unemployment_rate + 1.96 * ratio_unemployment_rate_sd / sqrt(50), color = "Wealth Tax and g", fill = "#3366ff", lty = "twodash") +
  geom_ribbon(data = subset(df_scenario10, periods >= 50), aes(x = periods, ymin = ratio_unemployment_rate, ymax = ratio_unemployment_rate + 1.96 * ratio_unemployment_rate_sd / sqrt(50), color = "Wealth Tax and g", fill = "#FFAB4D", lty = "dotted") +
  geom_ribbon(data = subset(df_scenario11, periods >= 50), aes(x = periods, ymin = ratio_unemployment_rate, ymax = ratio_unemployment_rate + 1.96 * ratio_unemployment_rate_sd / sqrt(50), color = "Wealth Tax and g", fill = "#ff0000", lty = "dashed") +
  theme(legend.position = "bottom") +
  labs(x = "Time", y = "Unemployment rate", color = "", fill = "") +
  theme(panel.border = element_rect(color = "black", size = 0.5, fill = NA),
    axis.title.x = element_text(size = 13),
    axis.title.y = element_text(size = 13),
    axis.text.x = element_text(size = 12),
    axis.text.y = element_text(size = 12),
    panel.background = element_rect(fill = "white"),
    legend.key = element_blank(),
    legend.position = "none",
    legend.background = element_blank()) +
  scale_color_manual(values = c("#45ff66", "#EB52ff", "#3366ff", "#FFAB4D", "#ff0000"),
    guide = guide_legend(override.aes = list(linetype = c("dotted", "dashed", "twodash")))) +
  scale_fill_manual(values = c("#45ff66", "#EB52ff", "#3366ff", "#FFAB4D", "#ff0000")) +
  geom_vline(xintercept = 50, linetype = "dashed", alpha = 0.5)

```



```
# Apply the calculations to each scenario
scenarios_wealthtax <- lapply(scenarios_wealthtax, function(df) {
  df$ratio_cpi <- df$cpi_Med / df_baseline$cpi_Med

  df$corr_scenario <- cor(df$cpi_Med, df_baseline$cpi_Med)

  df$ratio_cpi_sd <- abs(df$ratio_cpi) * sqrt(
    (df$cpi_Mad / df$cpi_Med)^2 +
    (df_baseline$cpi_Mad / df_baseline$cpi_Med)^2 -
    2 * df$corr_scenario * (df$cpi_Mad / df$cpi_Med) * (df_baseline$cpi_Mad / df_baseline$cpi_Med)
  )

  df$ratio_cpi_ci <- 1.96 * (df$ratio_cpi_sd / sqrt(50))

  return(df)
})

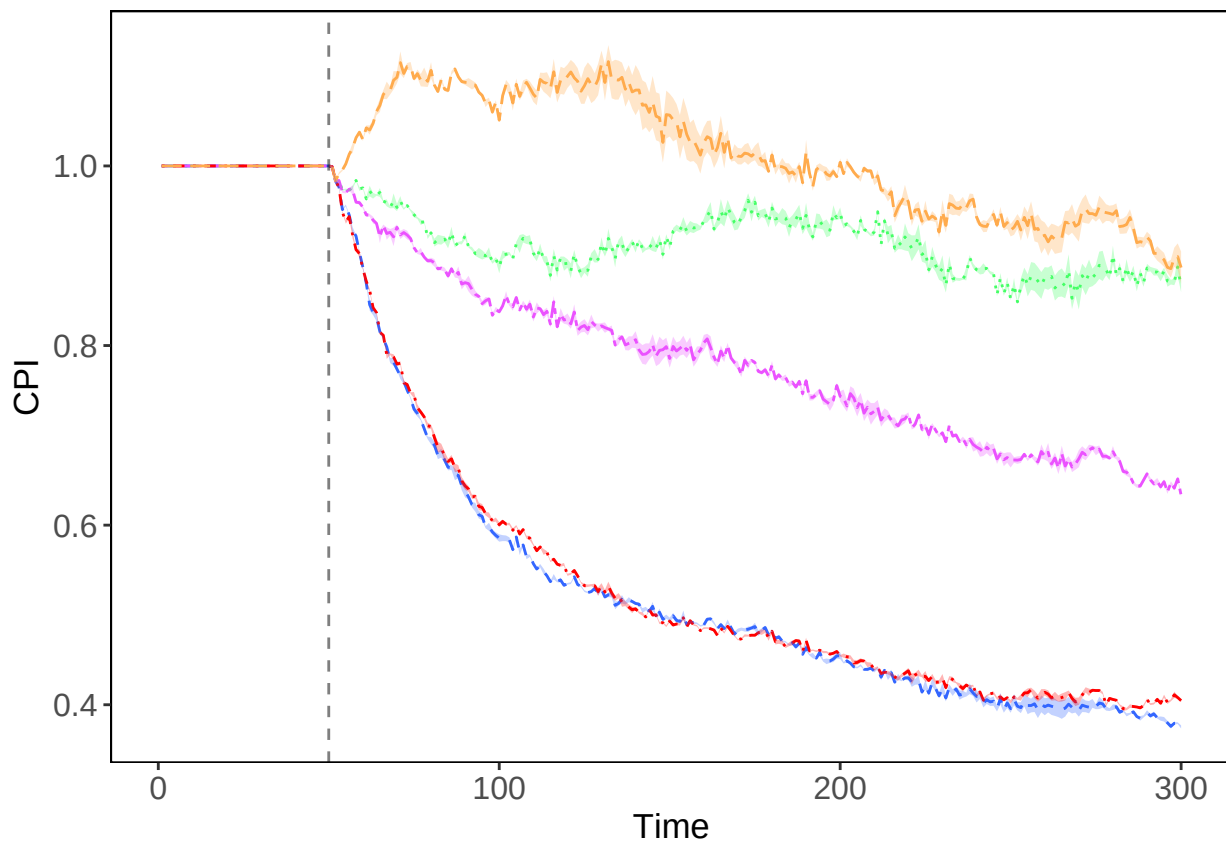
# Assign the updated data frames back to their respective variables
df_baseline <- scenarios_wealthtax[[1]]
df_scenario7 <- scenarios_wealthtax[[2]]
df_scenario8 <- scenarios_wealthtax[[3]]
df_scenario9 <- scenarios_wealthtax[[4]]
df_scenario10 <- scenarios_wealthtax[[5]]
df_scenario11 <- scenarios_wealthtax[[6]]

ggplot() +
```

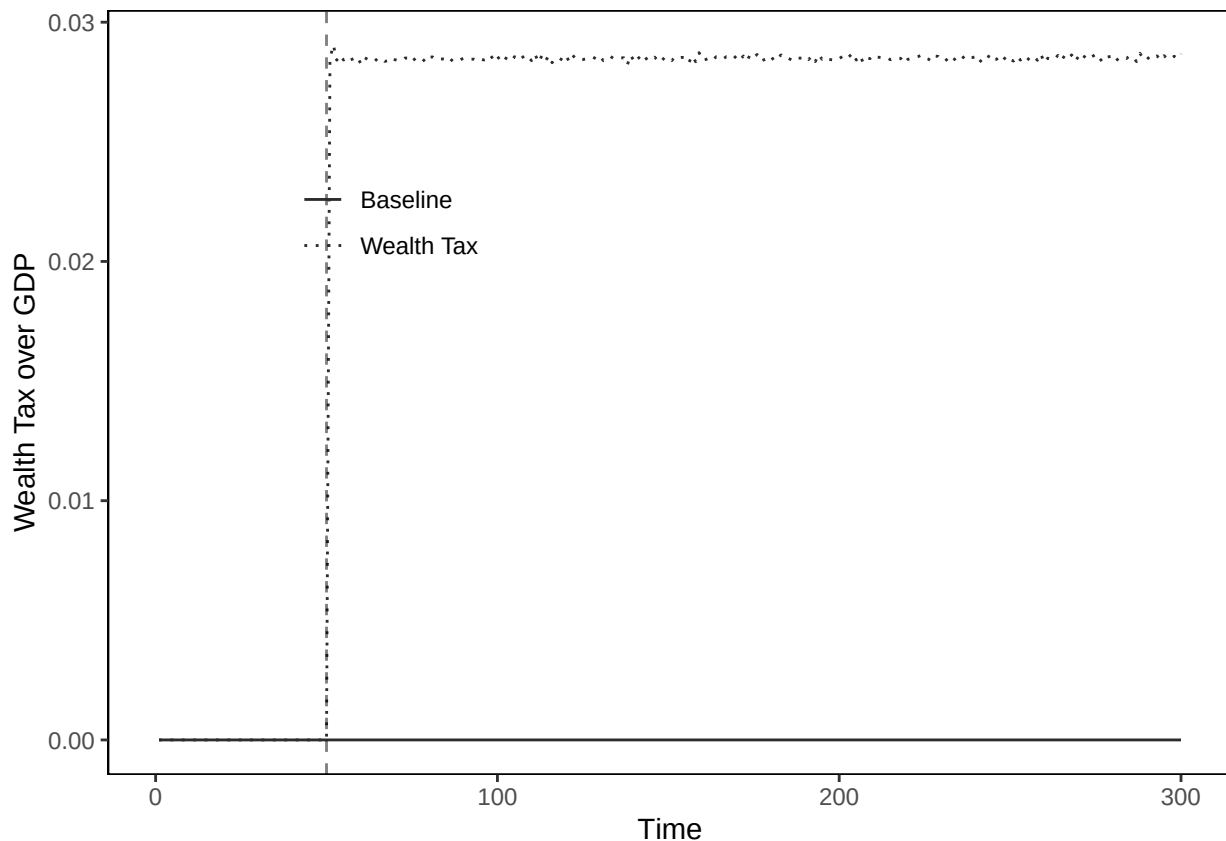
```

geom_line(data = df_scenario7, aes(x = periods, y = ratio_cpi, color = "Wealth Tax"), linetype = "dotted"),
geom_line(data = df_scenario8, aes(x = periods, y = ratio_cpi, color = "Wealth Tax and green public l"), linetype = "dotted"),
geom_line(data = df_scenario9, aes(x = periods, y = ratio_cpi, color = "Wealth Tax and green price su"), linetype = "dotted"),
geom_line(data = df_scenario10, aes(x = periods, y = ratio_cpi, color = "Wealth Tax and green research"), linetype = "dotted"),
geom_line(data = df_scenario11, aes(x = periods, y = ratio_cpi, color = "Wealth Tax and green public l"), linetype = "dotted"),
geom_ribbon(data = subset(df_scenario7, periods >= 50), aes(x = periods, ymin = ratio_cpi - ratio_cpi_50, ymax = ratio_cpi + ratio_cpi_50, color = "Wealth Tax"), linetype = "dotted"),
geom_ribbon(data = subset(df_scenario8, periods >= 50), aes(x = periods, ymin = ratio_cpi - ratio_cpi_50, ymax = ratio_cpi + ratio_cpi_50, color = "Wealth Tax and green public l"), linetype = "dotted"),
geom_ribbon(data = subset(df_scenario9, periods >= 50), aes(x = periods, ymin = ratio_cpi - ratio_cpi_50, ymax = ratio_cpi + ratio_cpi_50, color = "Wealth Tax and green price su"), linetype = "dotted"),
geom_ribbon(data = subset(df_scenario10, periods >= 50), aes(x = periods, ymin = ratio_cpi - ratio_cpi_50, ymax = ratio_cpi + ratio_cpi_50, color = "Wealth Tax and green research"), linetype = "dotted"),
geom_ribbon(data = subset(df_scenario11, periods >= 50), aes(x = periods, ymin = ratio_cpi - ratio_cpi_50, ymax = ratio_cpi + ratio_cpi_50, color = "Wealth Tax and green public l"), linetype = "dotted"),
theme(legend.position = "bottom") +
labs(x = "Time", y = "CPI", color = "", fill = "") +
theme(panel.border = element_rect(color = "black", size = 0.5, fill = NA),
      axis.title.x = element_text(size = 13),
      axis.title.y = element_text(size = 13),
      axis.text.x = element_text(size = 12),
      axis.text.y = element_text(size = 12),
      panel.background = element_rect(fill = "white"),
      legend.key = element_blank(),
      legend.position = "none",
      legend.background = element_blank()) +
scale_color_manual(values = c("#45ff66", "#EB52ff", "#3366ff", "#FFAB4D", "#ff0000"),
                  guide = guide_legend(override.aes = list(linetype = c("dotted", "dashed", "twodash", "solid", "longdash")))) +
scale_fill_manual(values = c("#45ff66", "#EB52ff", "#3366ff", "#FFAB4D", "#ff0000")) +
geom_vline(xintercept = 50, linetype = "dashed", alpha = 0.5)

```



```
ggplot() +
  geom_line(data = df_scenario7, aes(x = periods, y = wealth_tax_to_gdp_Med, color = "Wealth Tax"), linetype = "dotted") +
  geom_line(data = df_baseline, aes(x = periods, y = wealth_tax_to_gdp_Med, color = "Baseline"), alpha = 0.5) +
  theme(legend.position = "bottom") +
  labs(x = "Time", y = "Wealth Tax over GDP", color = "") +
  theme(panel.border = element_rect(color = "black", size = 0.5, fill = NA),
        panel.background = element_rect(fill = "white"),
        legend.key = element_blank(),
        legend.position = c(0.25, 0.75),
        legend.background = element_blank()) +
  scale_color_manual(values = c("#2c2c2c", "#2c2c2c"),
                    guide = guide_legend(override.aes = list(linetype = c("solid", "dotted")))) +
  geom_vline(xintercept = 50, linetype = "dashed", alpha = 0.5)
```



```
df_scenario1 <- df_baseline

scenario_names <- paste0("df_scenario", 1:11)

# Função para extrair último valor de cada variável de interesse
extract_average_values <- function(df_name) {
  df <- get(df_name)

  data.frame(
    scenario = df_name,
    emissions_coefficient_Med = mean(df$emissions_coefficient_Med, na.rm = TRUE),
    gdp_real_Med = mean(df$gdp_real_Med, na.rm = TRUE),
    emissions_Med = mean(df$emissions_Med, na.rm = TRUE),
```



```

    cpi_Med = mean(df$cpi_Med, na.rm = TRUE),
    unemployment_rate_Med = mean(df$unemployment_rate_Med, na.rm = TRUE),
    wage_share_Med = mean(df$wage_share_Med, na.rm = TRUE),
    gini_hh_Med = mean(df$gini_hh_Med, na.rm = TRUE)
  )
}

# Aplicar a função em todos os dataframes
summary_df <- map_dfr(scenario_names, extract_average_values)

# Transformar em formato longo para ordenar por variável
ranking_df <- summary_df %>%
  pivot_longer(-scenario, names_to = "variable", values_to = "value") %>%
  group_by(variable) %>%
  arrange(desc(value), .by_group = TRUE) %>%
  mutate(rank = row_number()) %>%
  ungroup()

# Voltar para formato wide com rankings por variável
ranking_wide <- ranking_df %>%
  select(scenario, variable, rank) %>%
  pivot_wider(names_from = variable, values_from = rank)

# Exibir
print(ranking_wide)

## # A tibble: 11 x 8
##   scenario      cpi_Med emissions_Med emissions_coefficient_Med gdp_real_Med
##   <chr>          <int>         <int>                  <int>         <int>
## 1 df_scenario6      1             7                      8             7
## 2 df_scenario2      2            10                      7            11
## 3 df_scenario4      3             4                      6             3
## 4 df_scenario3      4            11                     11             8
## 5 df_scenario5      5             8                      9             9
## 6 df_scenario1      6             2                      1             5
## 7 df_scenario11     7             5                      4             6
## 8 df_scenario7      8             6                      2            10
## 9 df_scenario9      9             1                      3             1
## 10 df_scenario10    10             3                      5             2
## 11 df_scenario8     11             9                     10             4
## # i 3 more variables: gini_hh_Med <int>, unemployment_rate_Med <int>,
## #   wage_share_Med <int>

write_xlsx(ranking_wide, "ranking.xlsx")

ranking_wide$scenario_name <- case_when(
  ranking_wide$scenario == "df_scenario1" ~ "BAU",
  ranking_wide$scenario == "df_scenario2" ~ "CT",
  ranking_wide$scenario == "df_scenario3" ~ "CT + RPL",
  ranking_wide$scenario == "df_scenario4" ~ "CT + PS",
  ranking_wide$scenario == "df_scenario5" ~ "CT + RS",
  ranking_wide$scenario == "df_scenario6" ~ "CT + PP",
  ranking_wide$scenario == "df_scenario7" ~ "WT",
  ranking_wide$scenario == "df_scenario8" ~ "WT + RLP",

```

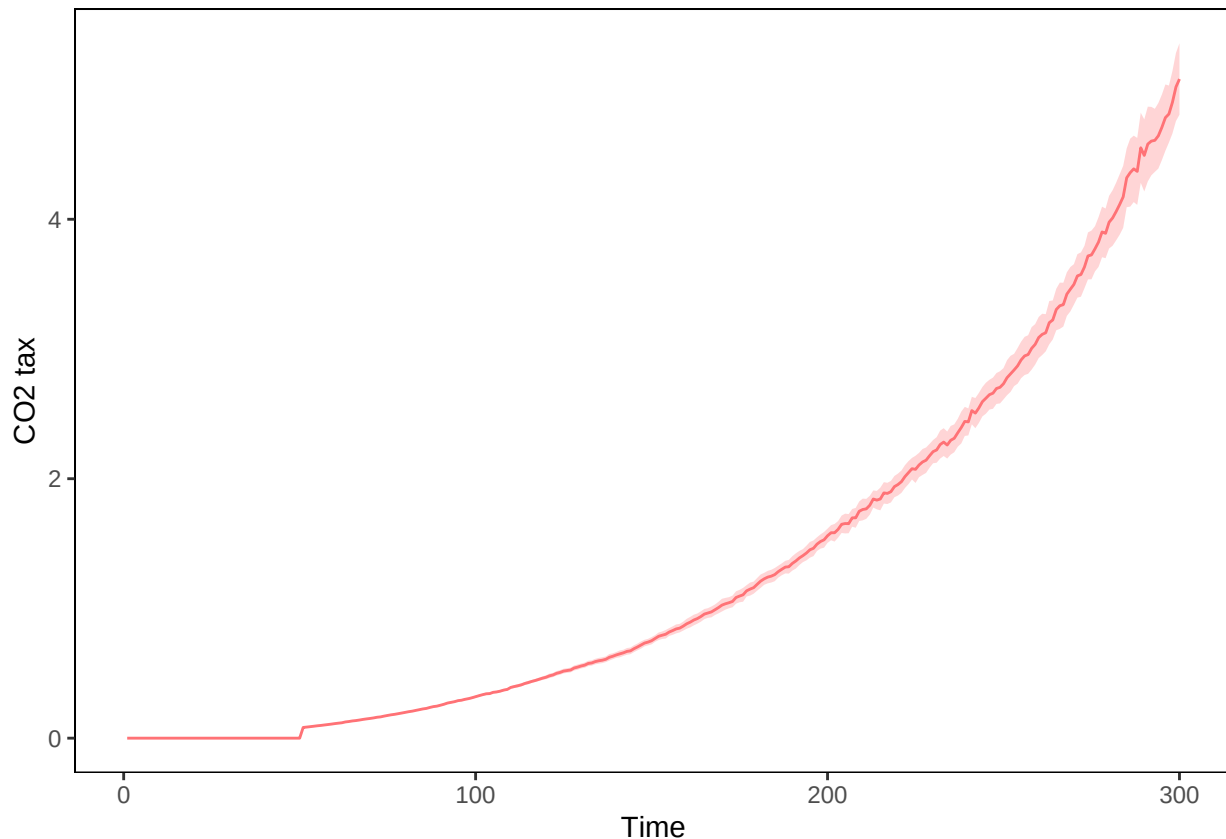


```

ranking_wide$scenario == "df_scenario9" ~ "WT + PS",
ranking_wide$scenario == "df_scenario10" ~ "WT + RS",
ranking_wide$scenario == "df_scenario11" ~ "WT + PP"
)

ggplot() +
  geom_line(data = df_scenario2, aes(x = periods, y = CO2_tax_Med, color = "Baseline"), alpha = 1) +
  geom_ribbon(data = df_scenario2, aes(x = periods, ymin = CO2_tax_Med - 1.96 * CO2_tax_Mad / sqrt(50),
                                     ymax = CO2_tax_Med + 1.96 * CO2_tax_Mad / sqrt(50), fill = "Baseline",
                                     alpha = 0.3, show.legend = FALSE) +
  scale_color_manual(values = c("Baseline" = "#FF7276")) +
  scale_fill_manual(values = c("Baseline" = "#FF7276")) +
  labs(x = "Time", y = "CO2 tax", color = "", fill = "") +
  theme(
    legend.position = "none",
    panel.border = element_rect(color = "black", size = 0.5, fill = NA),
    panel.background = element_rect(fill = "white")
  )

```

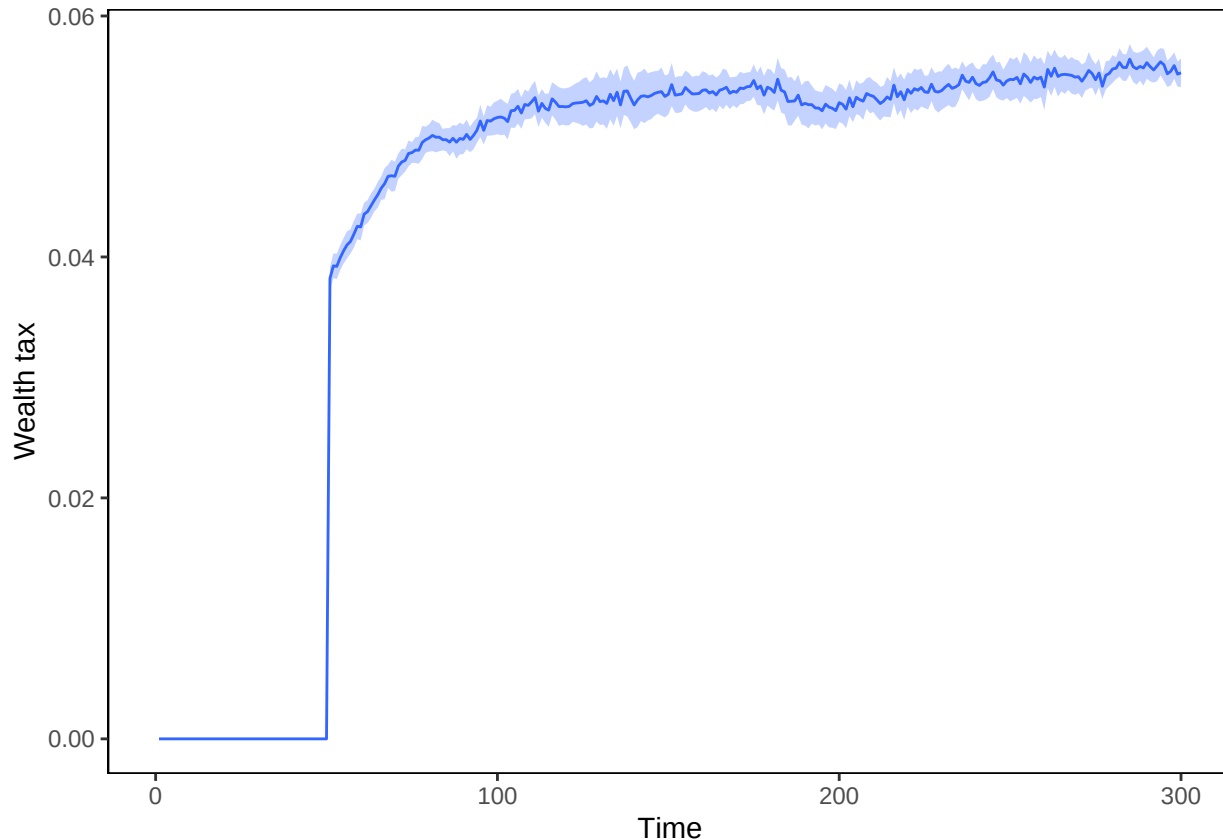


```

ggplot() +
  geom_line(data = df_scenario7, aes(x = periods, y = wealth_tax_rate_Med, color = "Baseline"), alpha = 1) +
  geom_ribbon(data = df_scenario7, aes(x = periods, ymin = wealth_tax_rate_Med - 1.96 * wealth_tax_rate_Mad / sqrt(50),
                                     ymax = wealth_tax_rate_Med + 1.96 * wealth_tax_rate_Mad / sqrt(50), fill = "Baseline",
                                     alpha = 0.3, show.legend = FALSE) +
  scale_color_manual(values = c("Baseline" = "#3366ff")) +
  scale_fill_manual(values = c("Baseline" = "#3366ff")) +
  labs(x = "Time", y = "Wealth tax", color = "", fill = "") +

```

```
theme(
  legend.position = "none",
  panel.border = element_rect(color = "black", size = 0.5, fill = NA),
  panel.background = element_rect(fill = "white")
)
```



```
ggplot() +
  geom_line(data = df_scenario2, aes(x = periods, y = ratio_cpi, color = "Carbon Tax"), linetype = "dotted") +
  geom_line(data = df_scenario3, aes(x = periods, y = ratio_cpi, color = "Carbon Tax and green public l"), linetype = "dotted") +
  geom_line(data = df_scenario4, aes(x = periods, y = ratio_cpi, color = "Carbon Tax and green price su"), linetype = "dotted") +
  geom_line(data = df_scenario5, aes(x = periods, y = ratio_cpi, color = "Carbon Tax and green research"), linetype = "dotted") +
  geom_line(data = df_scenario6, aes(x = periods, y = ratio_cpi, color = "Carbon Tax and green public p"), linetype = "dotted") +
  theme(legend.position = "bottom") +
  labs(x = "", y = "", color = "", fill = "") +
  theme(panel.border = element_rect(color = "black", size = 0, fill = NA),
        axis.text.x = element_text(size = 0),
        axis.text.y = element_text(size = 0),
        legend.key.width = unit(4, "cm"), # make line long
        legend.key.height = unit(0.6, "cm"), # keep it thin
        panel.background = element_rect(fill = "white"),
        legend.key = element_blank(),
        legend.position = c(0.5, 0.5),
        axis.ticks = element_blank(),
        legend.text = element_text(size = 14),
        legend.spacing.y = unit(2, "cm"),
        legend.background = element_blank()) +
  scale_color_manual(values = c("#45ff66", "#EB52ff", "#3366ff", "#FFAB4D", "#ff0000"),
```

```

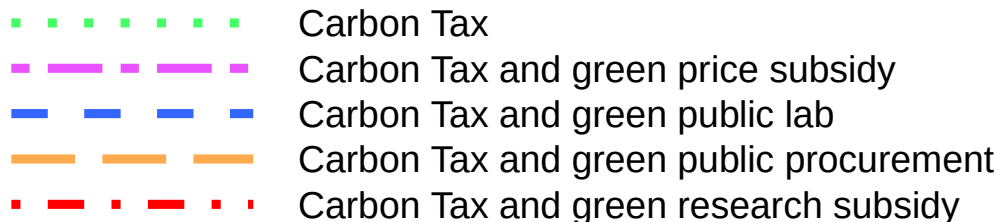
        guide = guide_legend(override.aes = list(linetype = c("dotted", "dashed", "twodash", "solid"),
scale_fill_manual(values = c("#45ff66", "#EB52ff", "#3366ff", "#FFAB4D", "#ff0000")) +
scale_y_continuous(limits = c(10, 100))+
guides(color = guide_legend(
  override.aes = list(linewidth = 1.6)
))

```

```

## Warning: Removed 300 rows containing missing values or values outside the scale range
## (`geom_line()`).
## Removed 300 rows containing missing values or values outside the scale range
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## (`geom_line()`).

```



■ ■ ■ ■ ■ ■ ■ Carbon Tax
- - - - - Carbon Tax and green price subsidy
- - - - - Carbon Tax and green public lab
- - - - - Carbon Tax and green public procurement
■ ■ ■ ■ ■ ■ ■ Carbon Tax and green research subsidy

```

ggplot() +
  geom_line(data = df_scenario7, aes(x = periods, y = ratio_cpi, color = "Wealth Tax"), linetype = "dotted",
  geom_line(data = df_scenario8, aes(x = periods, y = ratio_cpi, color = "Wealth Tax and green public lab"), linetype = "dashed",
  geom_line(data = df_scenario9, aes(x = periods, y = ratio_cpi, color = "Wealth Tax and green price subsidy"), linetype = "twodash",
  geom_line(data = df_scenario10, aes(x = periods, y = ratio_cpi, color = "Wealth Tax and green research subsidy"), linetype = "solid",
  geom_line(data = df_scenario11, aes(x = periods, y = ratio_cpi, color = "Wealth Tax and green public lab"), linetype = "dotted",
  theme(legend.position = "bottom") +
  labs(x = "", y = "", color = "", fill = "") +
  theme(panel.border = element_rect(color = "black", size = 0, fill = NA),
        axis.text.x = element_text(size = 0),
        axis.text.y = element_text(size = 0),

```

```

legend.key.width = unit(4, "cm"), # make line long
legend.key.height = unit(0.6, "cm"), # keep it thin
panel.background = element_rect(fill = "white"),
legend.key = element_blank(),
legend.position = c(0.5, 0.5),
axis.ticks = element_blank(),
legend.text = element_text(size = 14),
legend.spacing.y = unit(2, "cm"),
legend.background = element_blank() +
scale_color_manual(values = c("#45ff66", "#EB52ff", "#3366ff", "#FFAB4D", "#ff0000"),
                    guide = guide_legend(override.aes = list(linetype = c("dotted", "dashed", "twodash", "solid", "longdash"),
scale_fill_manual(values = c("#45ff66", "#EB52ff", "#3366ff", "#FFAB4D", "#ff0000")) +
scale_y_continuous(limits = c(1.5, 100))+
guides(color = guide_legend(
  override.aes = list(linewidth = 1.6)
))

```

```

## Warning: Removed 300 rows containing missing values or values outside the scale range
## (`geom_line()`).
## Removed 300 rows containing missing values or values outside the scale range
## (`geom_line()`).
## Removed 300 rows containing missing values or values outside the scale range
## (`geom_line()`).
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## (`geom_line()`).
## Removed 300 rows containing missing values or values outside the scale range
## (`geom_line()`).

```

■ ■ ■ ■ ■ ■ ■ ■	Wealth Tax
■ ■ ■ ■ ■ ■ ■ ■	Wealth Tax and green price subsidy
■ ■ ■ ■ ■ ■ ■ ■	Wealth Tax and green public lab
■ ■ ■ ■ ■ ■ ■ ■	Wealth Tax and green public procurement
■ ■ ■ ■ ■ ■ ■ ■	Wealth Tax and green research subsidy